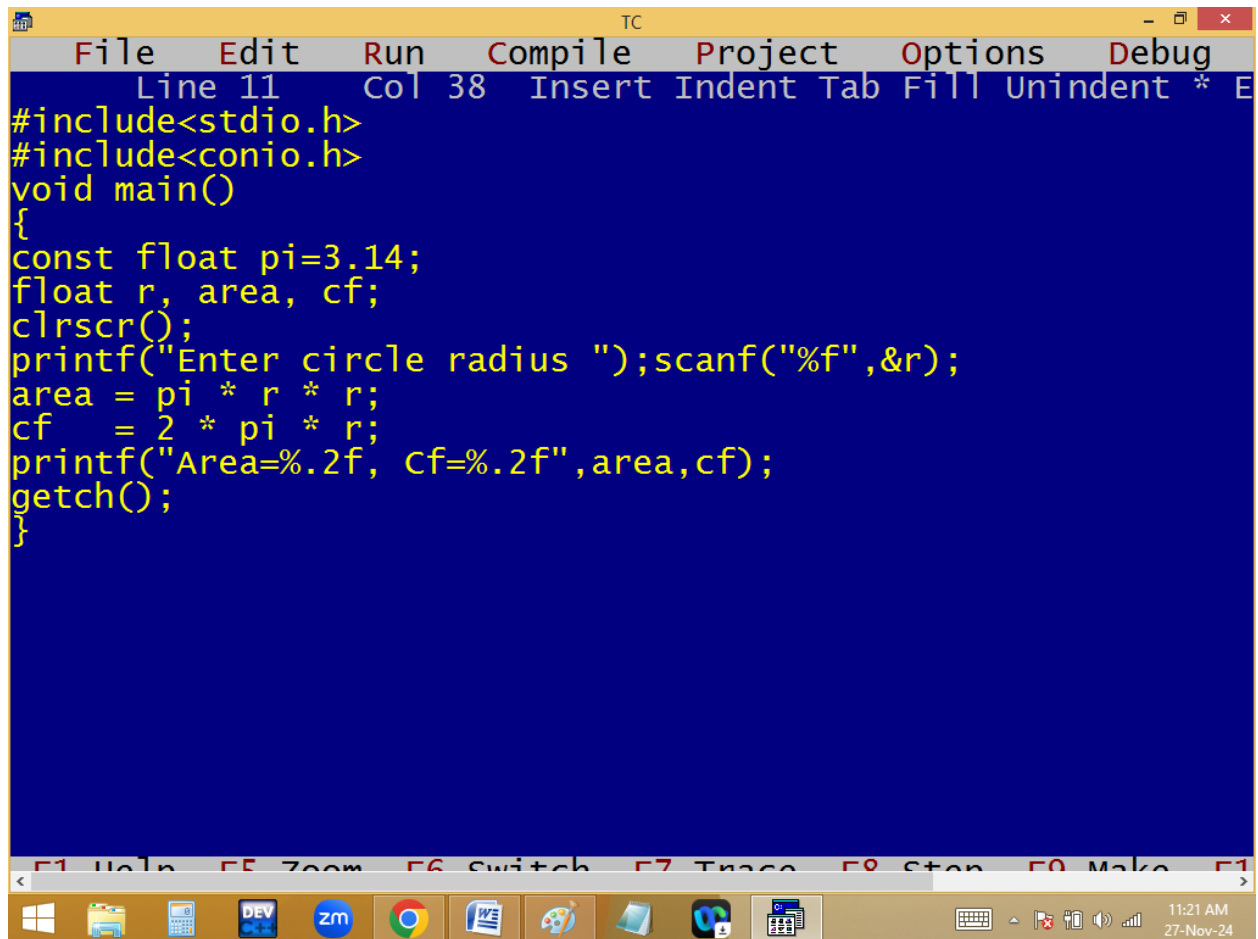


Read the radius of a circle and find the area and circumference.

$$\text{Area} = \pi * r * r$$

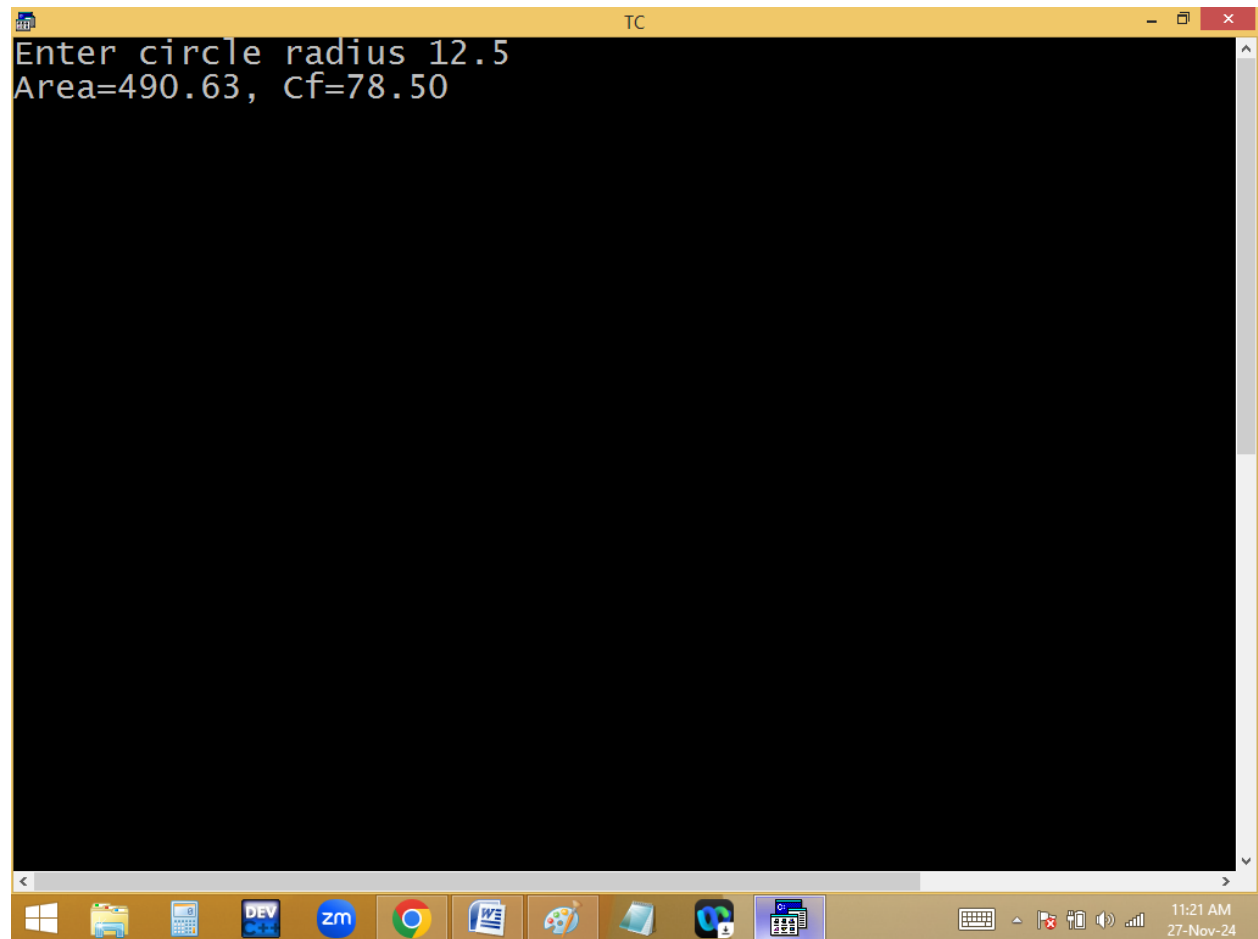
$$Cf = 2 * \pi * r$$

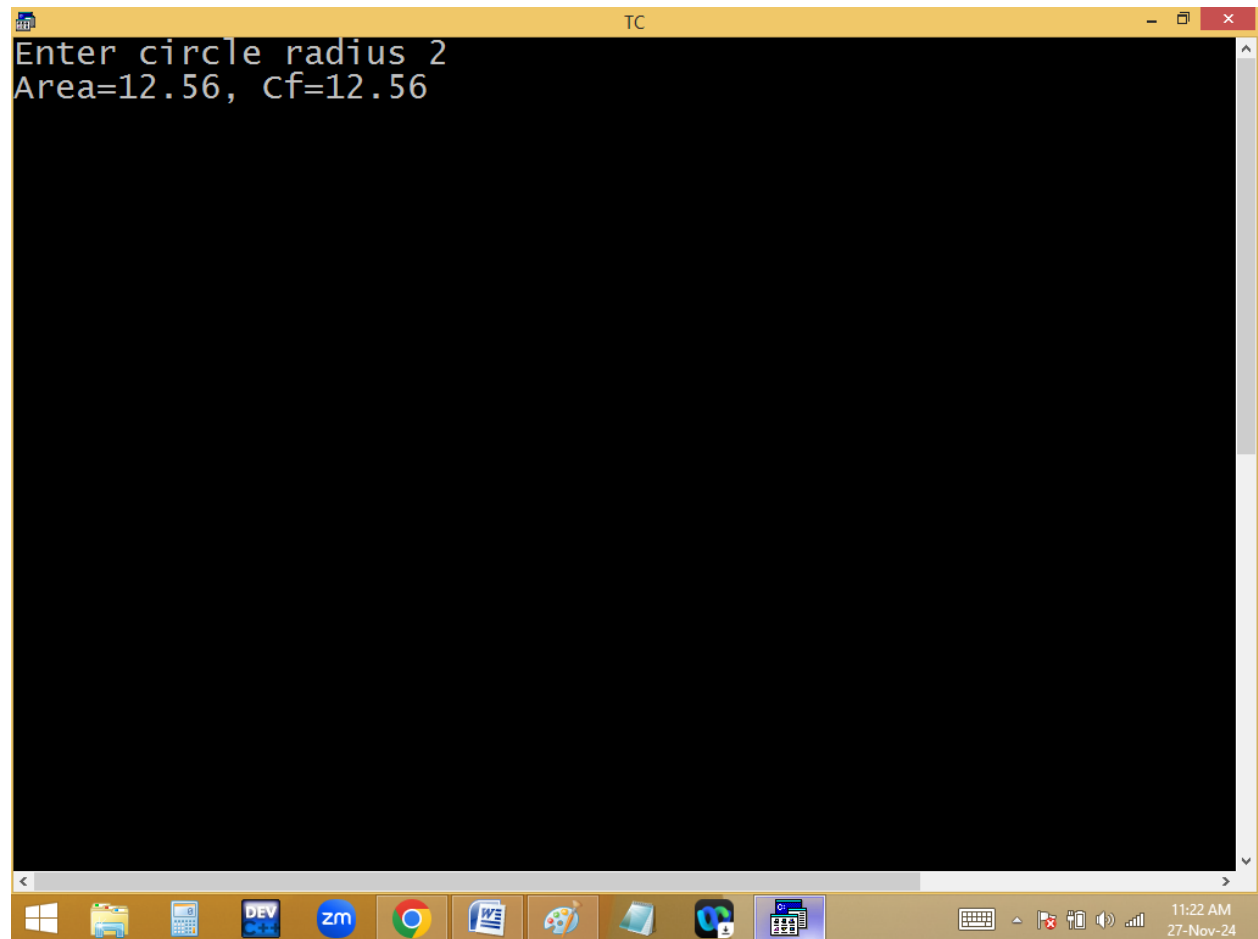


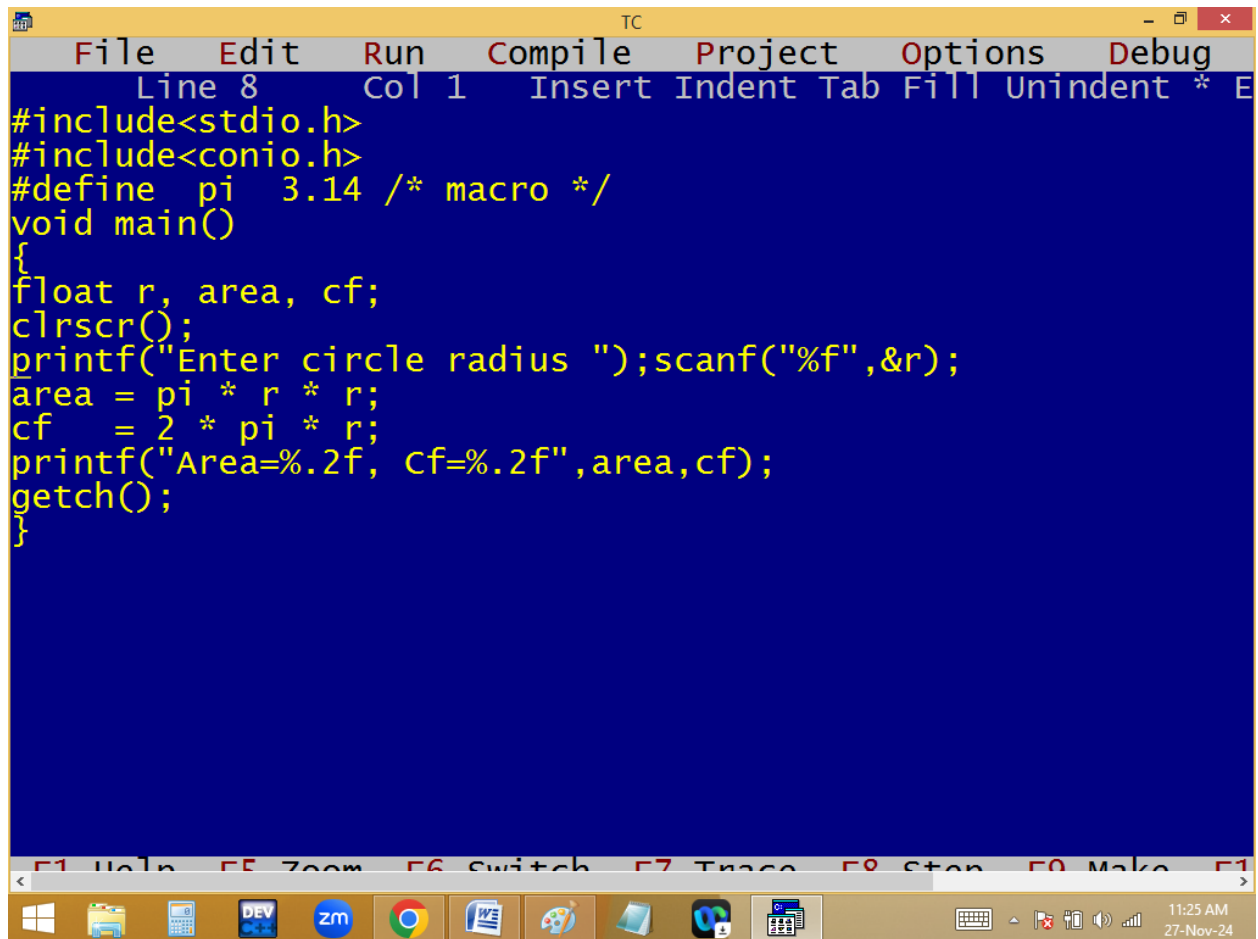
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 11", "Col 38", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
const float pi=3.14;
float r, area, cf;
clrscr();
printf("Enter circle radius ");scanf("%f",&r);
area = pi * r * r;
cf    = 2 * pi * r;
printf("Area=%.2f, Cf=%.2f",area,cf);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner shows "11:21 AM" and "27-Nov-24".



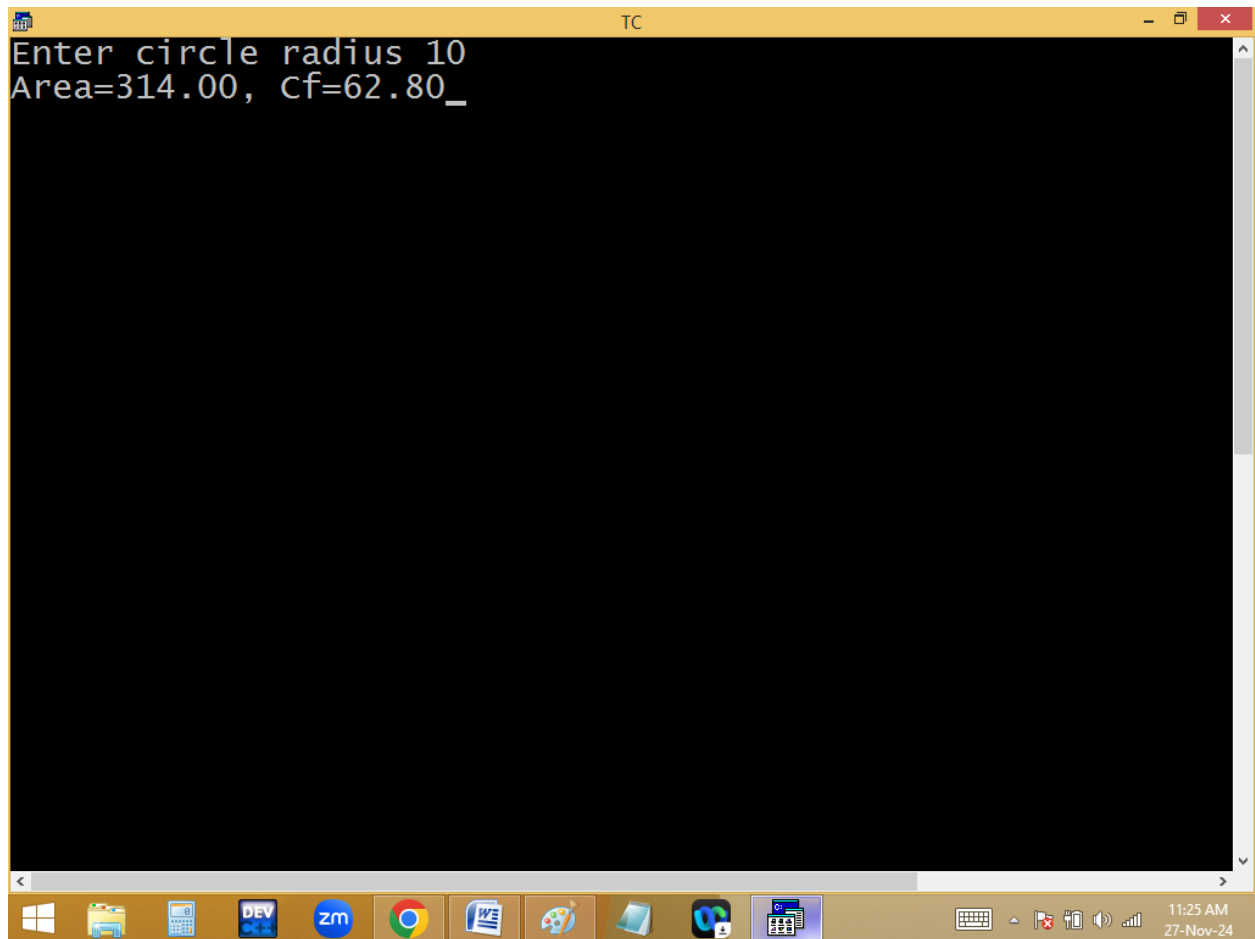




The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 8", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
#define pi 3.14 /* macro */
void main()
{
float r, area, cf;
clrscr();
printf("Enter circle radius ");scanf("%f",&r);
area = pi * r * r;
cf = 2 * pi * r;
printf("Area=%.2f, Cf=%.2f",area,cf);
getch();
}
```

Below the code, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner shows "11:25 AM" and "27-Nov-24".



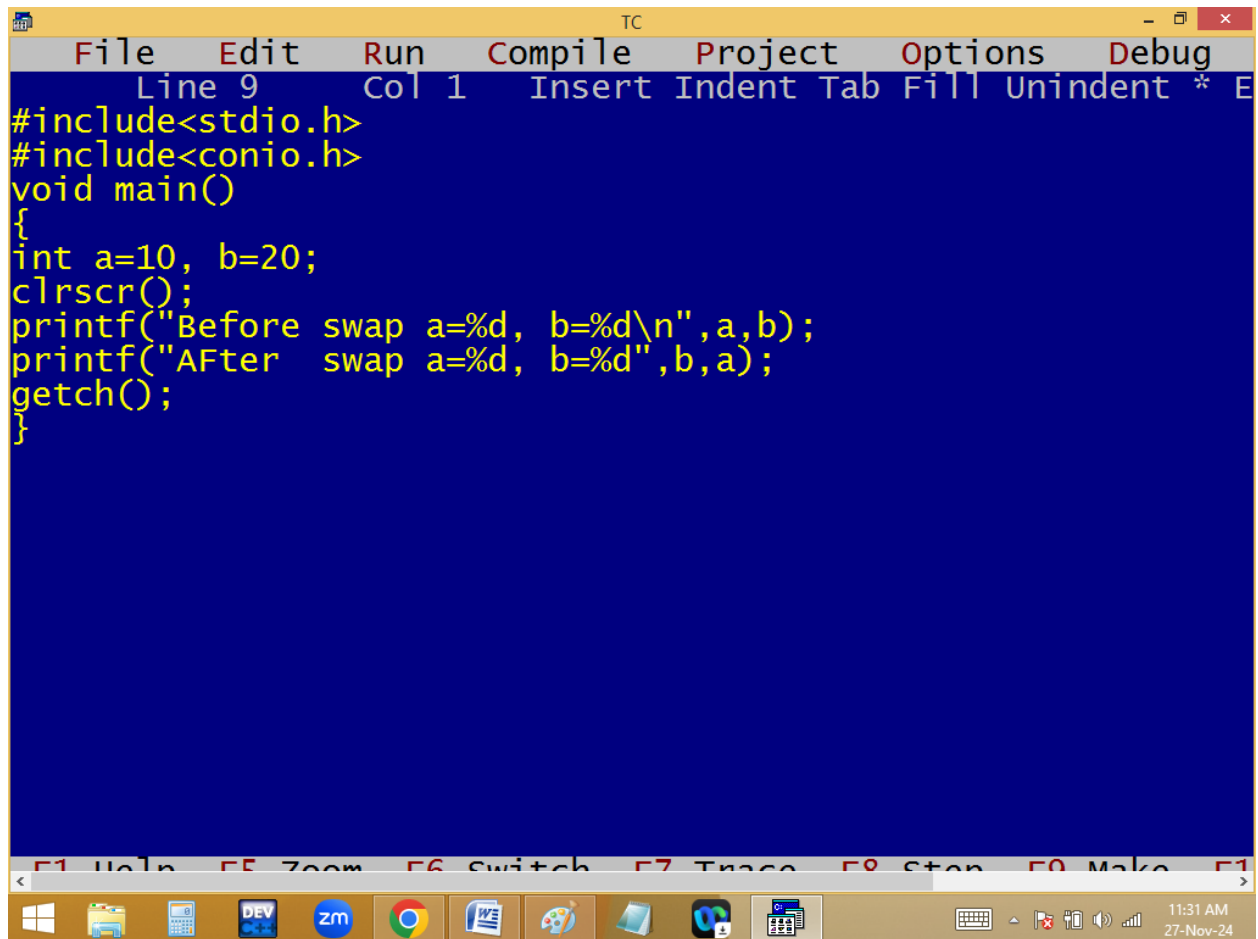
The screenshot shows a Turbo C++ (TC) window with a black background and white text. The text displays the input 'Enter circle radius 10' and the calculated output 'Area=314.00, Cf=62.80_'. The window title bar reads 'TC'. The Windows taskbar at the bottom includes icons for various applications and the system clock showing 11:25 AM on 27-Nov-24.

```
Enter circle radius 10
Area=314.00, Cf=62.80_
```

Read two numbers and swap [interchange] the values:

$a=10, b=20 \rightarrow a=20, b=10$

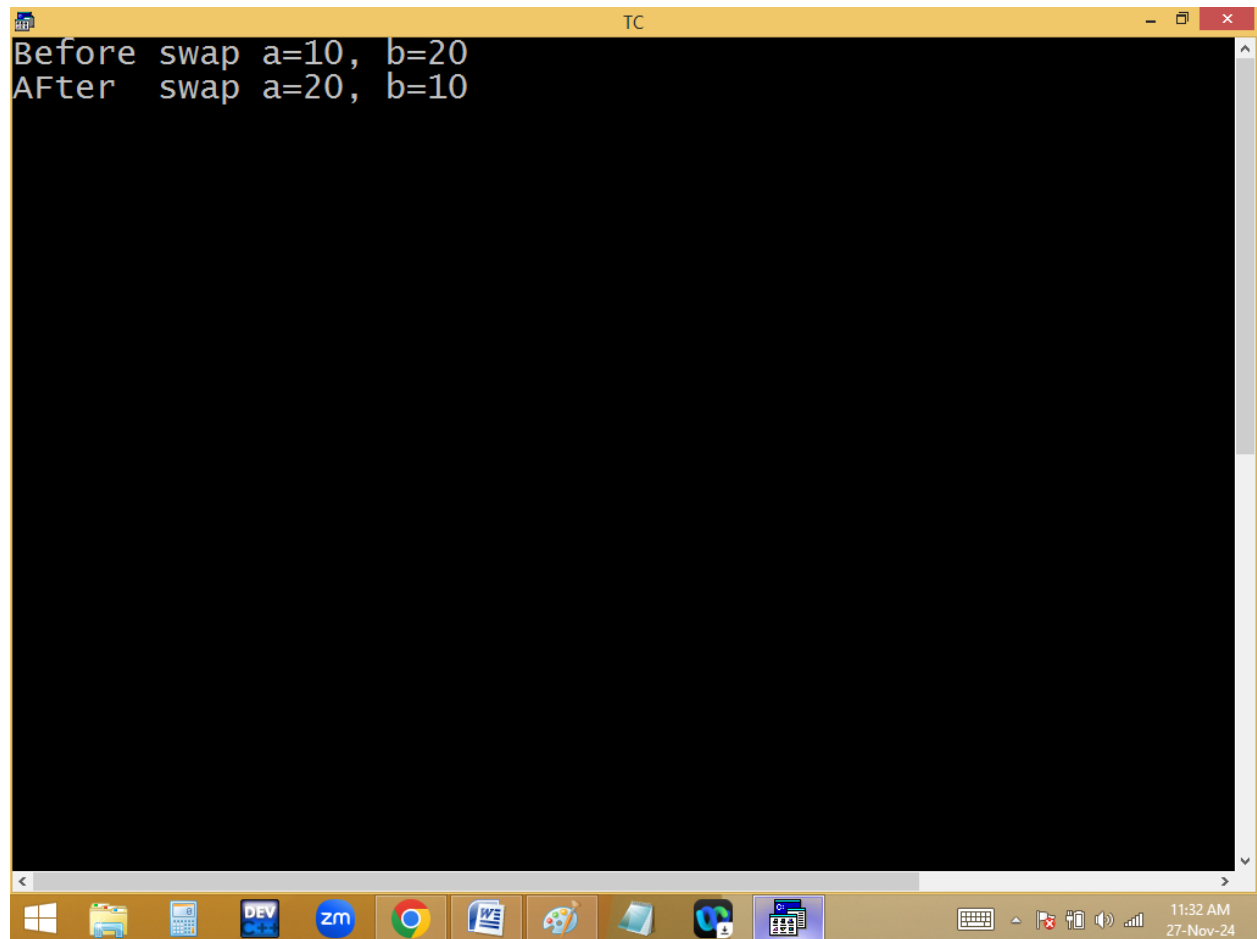
Without using operators:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 9", "Col 1", and "Insert Indent Tab Fill Unindent * E". The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=20;
clrscr();
printf("Before swap a=%d, b=%d\n",a,b);
printf("After swap a=%d, b=%d",b,a);
getch();
}
```

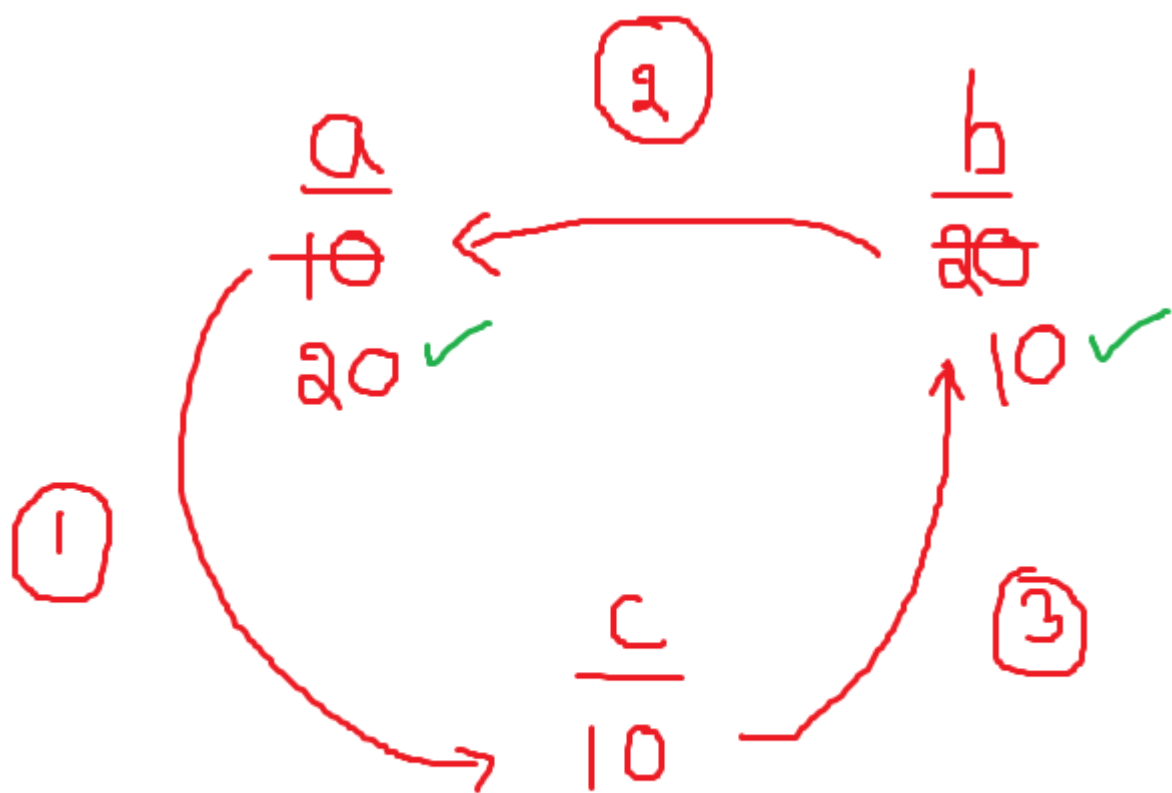
Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner shows "11:31 AM" and "27-Nov-24".

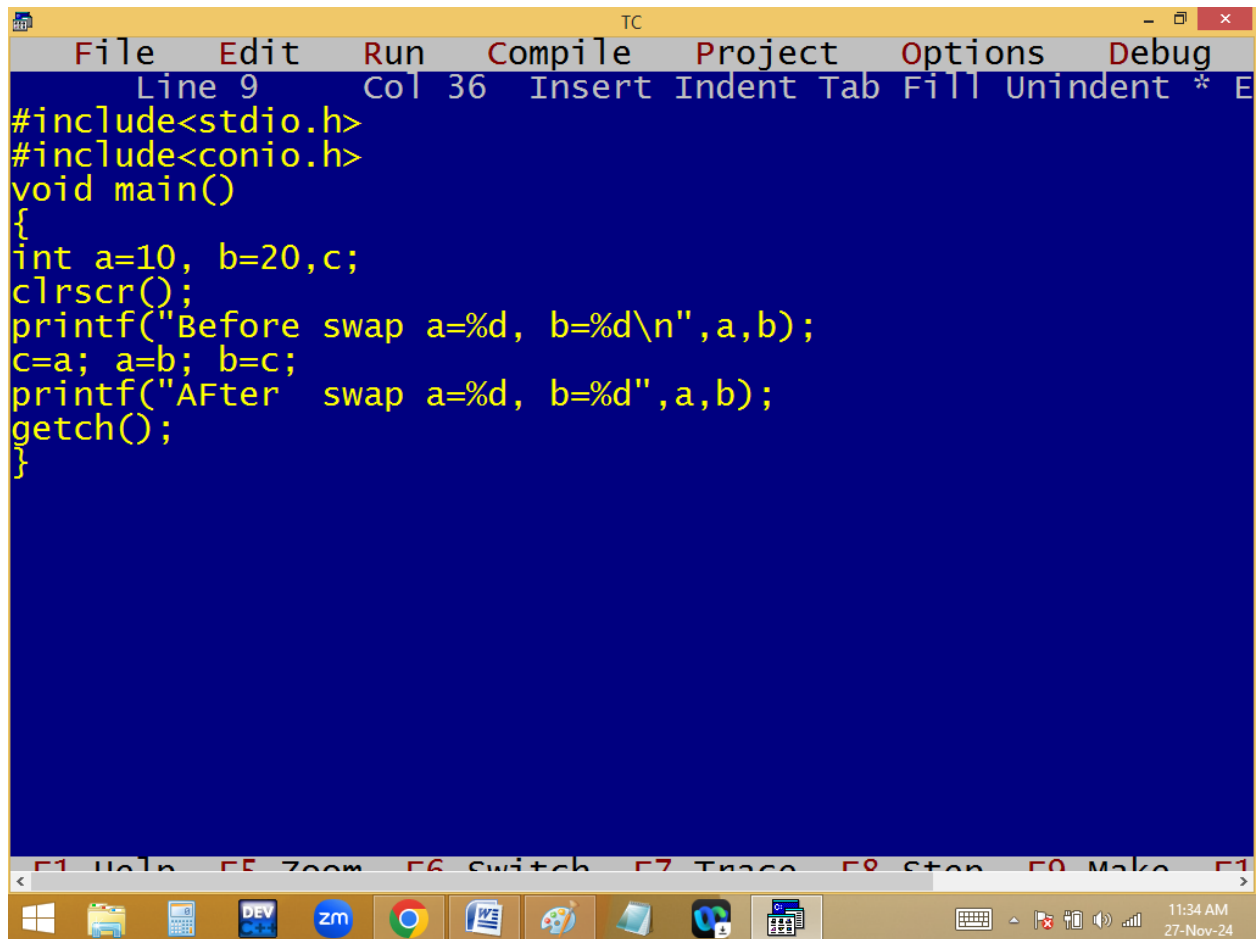


The image shows a screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" in the center. The main area is black with white text. The text reads: "Before swap a=10, b=20" on the first line and "After swap a=20, b=10" on the second line. The window has a standard Windows taskbar at the bottom with various icons including the Start button, File Explorer, Calculator, DEV, zm, Chrome, Word, Paint, and a folder icon. The system tray on the right shows the time as 11:32 AM and the date as 27-Nov-24.

```
Before swap a=10, b=20
After swap a=20, b=10
```

Using 3rd variable:



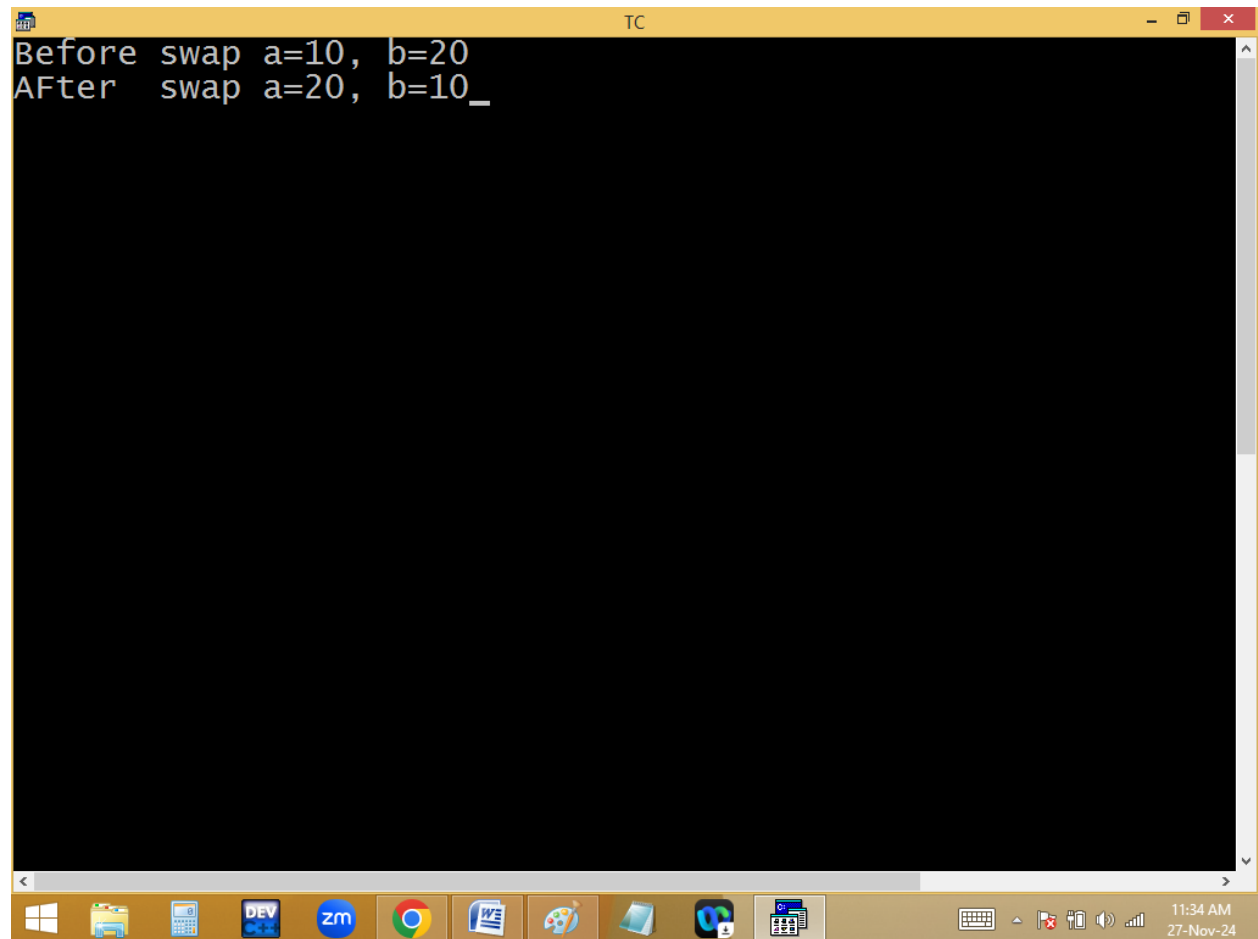


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 9", "Col 36", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

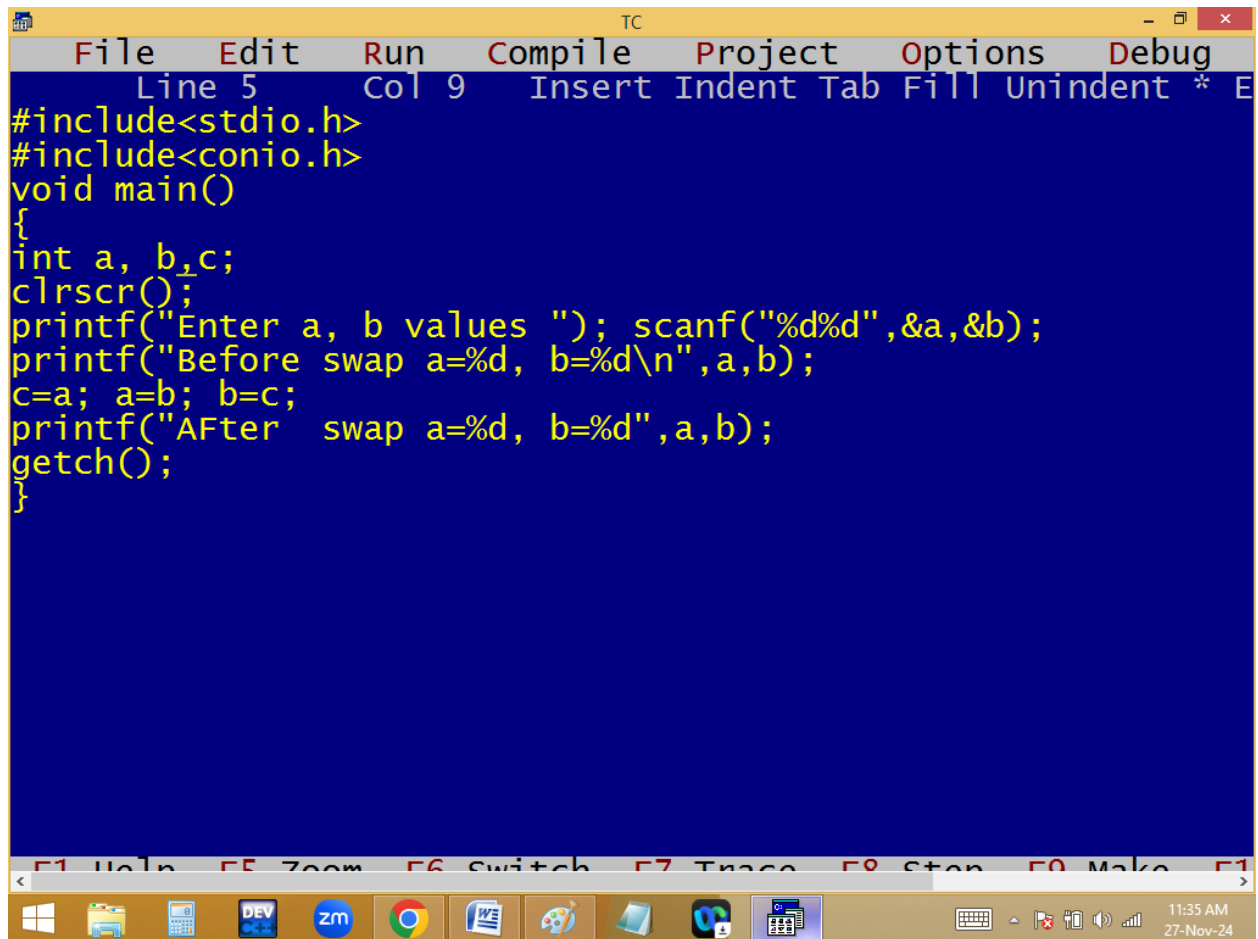
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=20,c;
clrscr();
printf("Before swap a=%d, b=%d\n",a,b);
c=a; a=b; b=c;
printf("After swap a=%d, b=%d",a,b);
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "11:34 AM" and the date "27-Nov-24".

```
TC
Before swap a=10, b=20
After swap a=20, b=10_
```



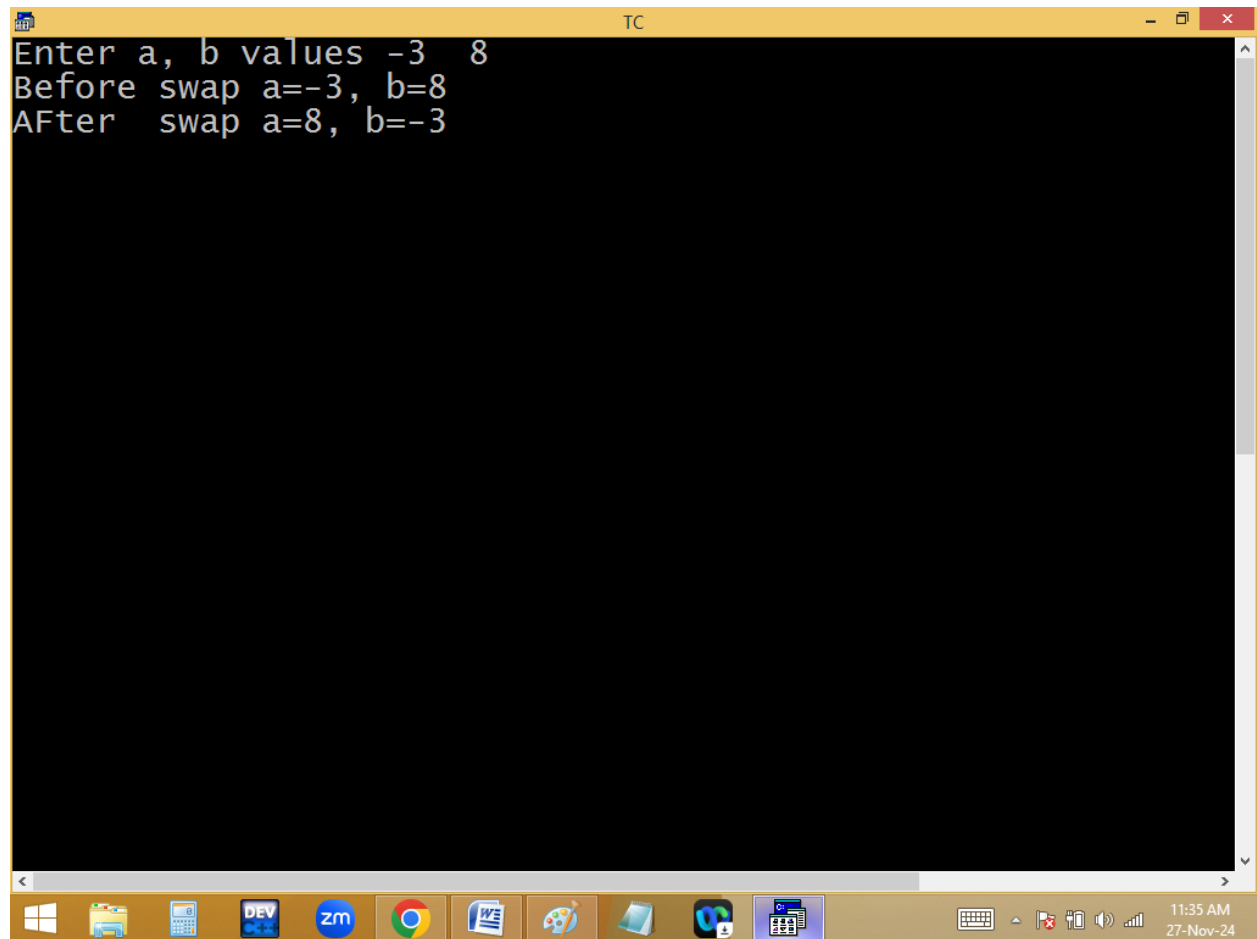
The image shows a Windows 10 desktop environment. A Turbo C++ (TC) window is open, displaying the output of a swap function. The text in the window reads: "Before swap a=10, b=20" and "After swap a=20, b=10_". The taskbar at the bottom contains several application icons, including the Windows Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Microsoft Word, a game controller icon, a folder icon, a game icon, and a calendar icon. The system tray on the right shows the time as 11:34 AM and the date as 27-Nov-24.



```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 9 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b, c;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
printf("Before swap a=%d, b=%d\n",a,b);
c=a; a=b; b=c;
printf("After swap a=%d, b=%d",a,b);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

11:35 AM
27-Nov-24



The image shows a screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" and standard minimize, maximize, and close buttons. The main area is black with white text. The text displayed is:

```
Enter a, b values -3 8  
Before swap a=-3, b=8  
After swap a=8, b=-3
```

At the bottom of the window is a Windows taskbar with various icons including the Start button, File Explorer, Calculator, DEV, zm, Chrome, Word, Paint, and a folder. On the right side of the taskbar, there are system icons for keyboard, network, and volume, along with the date and time: "11:35 AM 27-Nov-24".

Without using 3rd variable:

```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 19 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
printf("Before swap a=%d, b=%d\n",a,b);
/*a=a+b; b=a-b;a=a-b;
a=a*b; b=a/b; a=a/b;*/
a=a^b; b=a^b; a=a^b;
printf("After swap a=%d, b=%d",a,b);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

11:37 AM 27-Nov-24

```
TC
Enter a, b values 10 20
Before swap a=10, b=20
After swap a=20, b=10
```

a=10==>30==>20
b=20==>10

a=a+b==>10+20=30
b=a-b==>30-20=10
a=a-b==>30-10=20

a=10==>200==>20
b=20==>10

a=a*b==>10*20=200
b=a/b==>200/20=10
a=a/b==>200/10=20

$$\begin{array}{r}
 2 \overline{) 10} \\
 2 \overline{) 5-0} \\
 2 \overline{) 2-1} \\
 1-0
 \end{array}$$

$$\begin{array}{r}
 2 \overline{) 20} \\
 2 \overline{) 10-0} \\
 2 \overline{) 5-0} \\
 2 \overline{) 2-1} \\
 1-0
 \end{array}$$

$a = 1010$
 $b = 10100$

$a=10, b=20$

$a = a \wedge b \leftarrow$

$a=10=01010$

$b=20=\underline{10100}$

$11110=30$

$$\begin{array}{cccc}
 2^4 & 2^3 & 2^2 & 2^1 \\
 / & / & / & / \\
 16 & 8 & 4 & 2 \\
 + & + & + & + \\
 16 & 8 & 4 & 2 = 30
 \end{array}$$

$a=30, b=20$

$b=a \wedge b; \leftarrow$

$a=30=11110$

$b=20=\underline{10100}$

$01010=10$

$a=30, b=10$

$a=a \wedge b; \leftarrow$

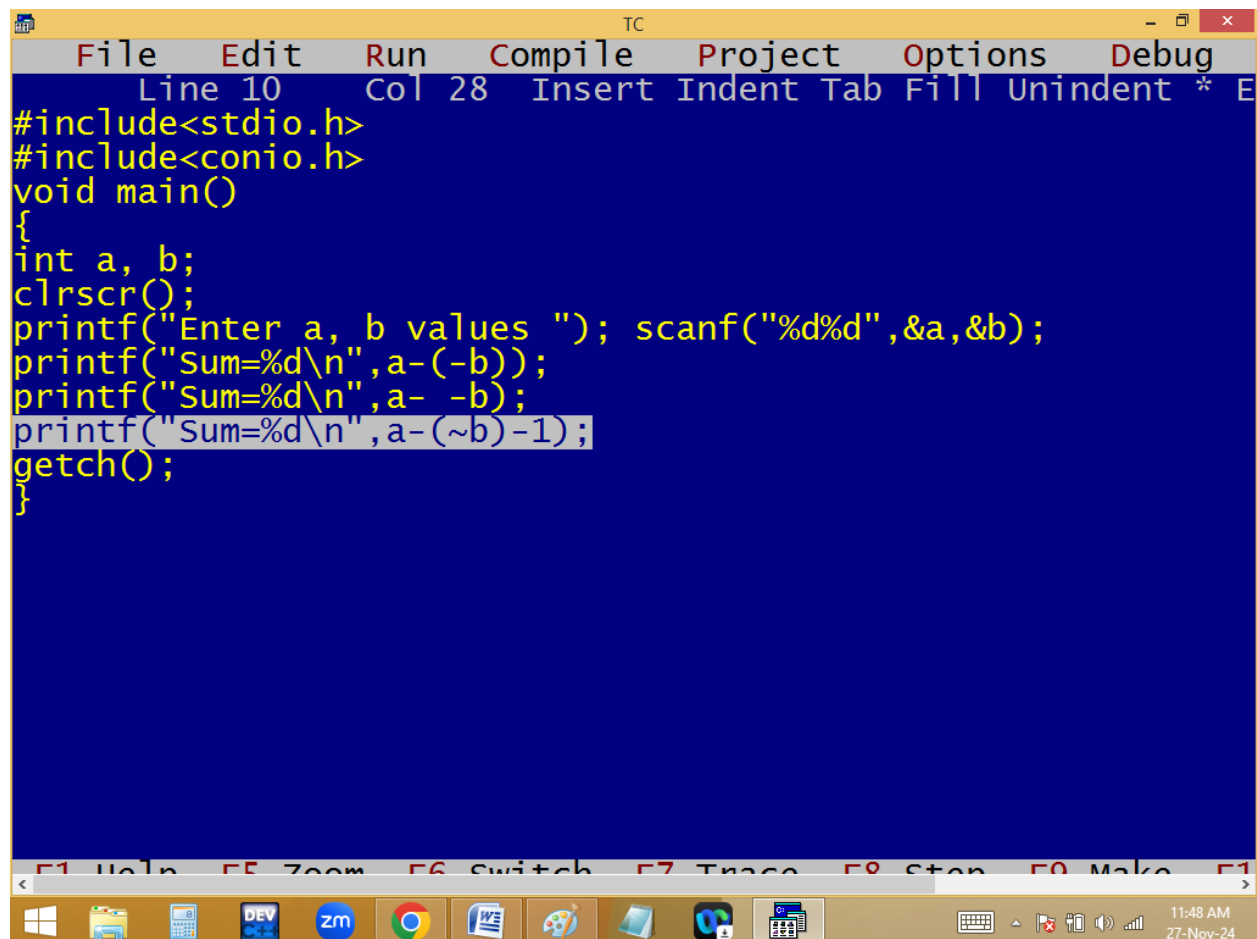
$a=30=11110$

$b=10=\underline{01010}$

$10100=20$

$a=20, b=10 \checkmark$

Add two numbers without using + operator?



The screenshot shows a Turbo C++ IDE window titled "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 10 Col 28 Insert Indent Tab Fill Unindent * E". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
printf("Sum=%d\n",a-(-b));
printf("Sum=%d\n",a- -b);
printf("Sum=%d\n",a-(~b)-1);
getch();
}
```

The third line of the printf statement, `printf("Sum=%d\n",a- -b);`, is highlighted. The bottom status bar shows function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar at the bottom displays various application icons and the system clock showing 11:48 AM on 27-Nov-24.

```
TC
Enter a, b values 10 20
Sum=30
Sum=30
Sum=30
_
```

$$a-(\sim b)-1$$

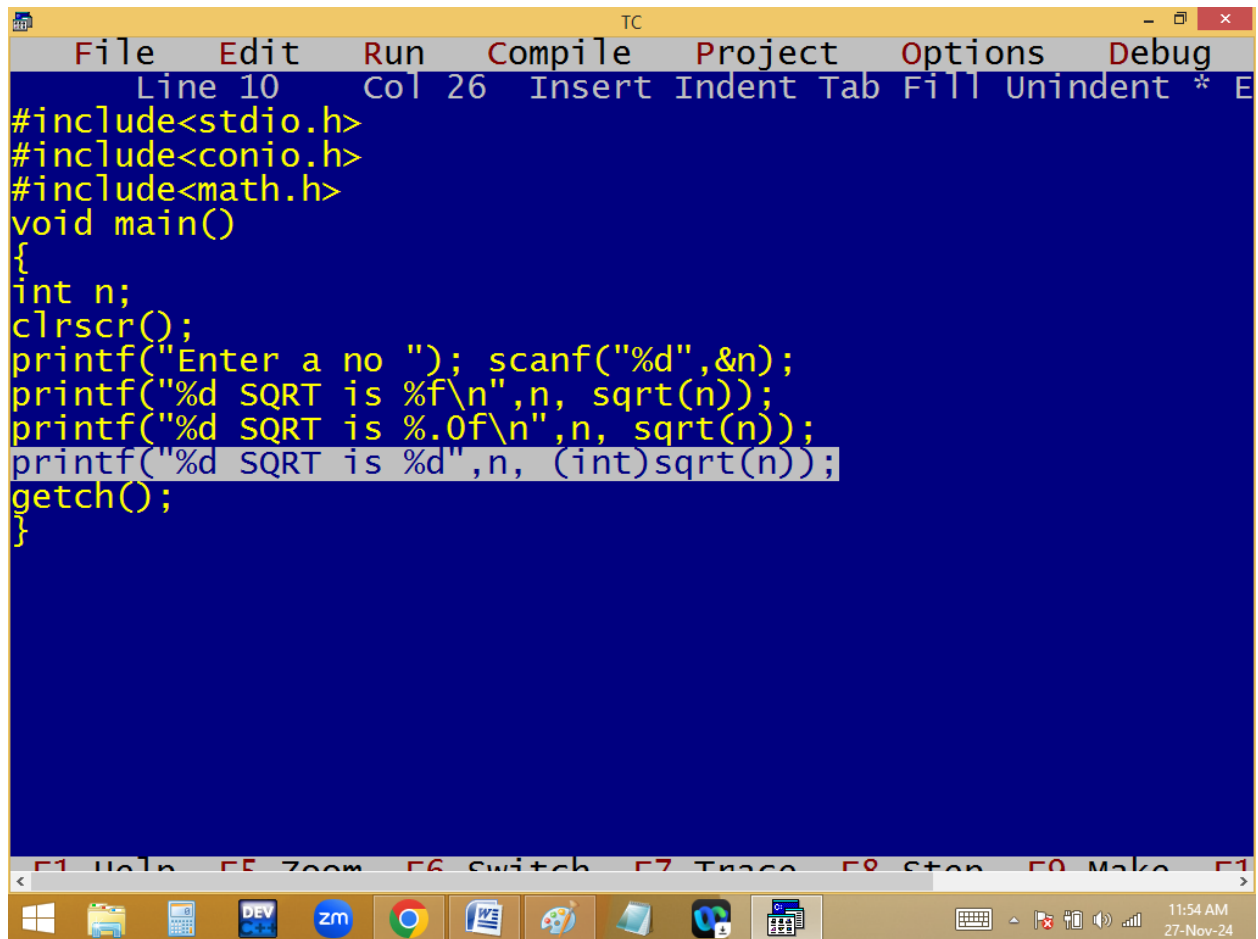
$$10-(\sim 20)-1$$

$$10-(-21)-1$$

$$10+21-1=30$$

Finding SQRT of given no?

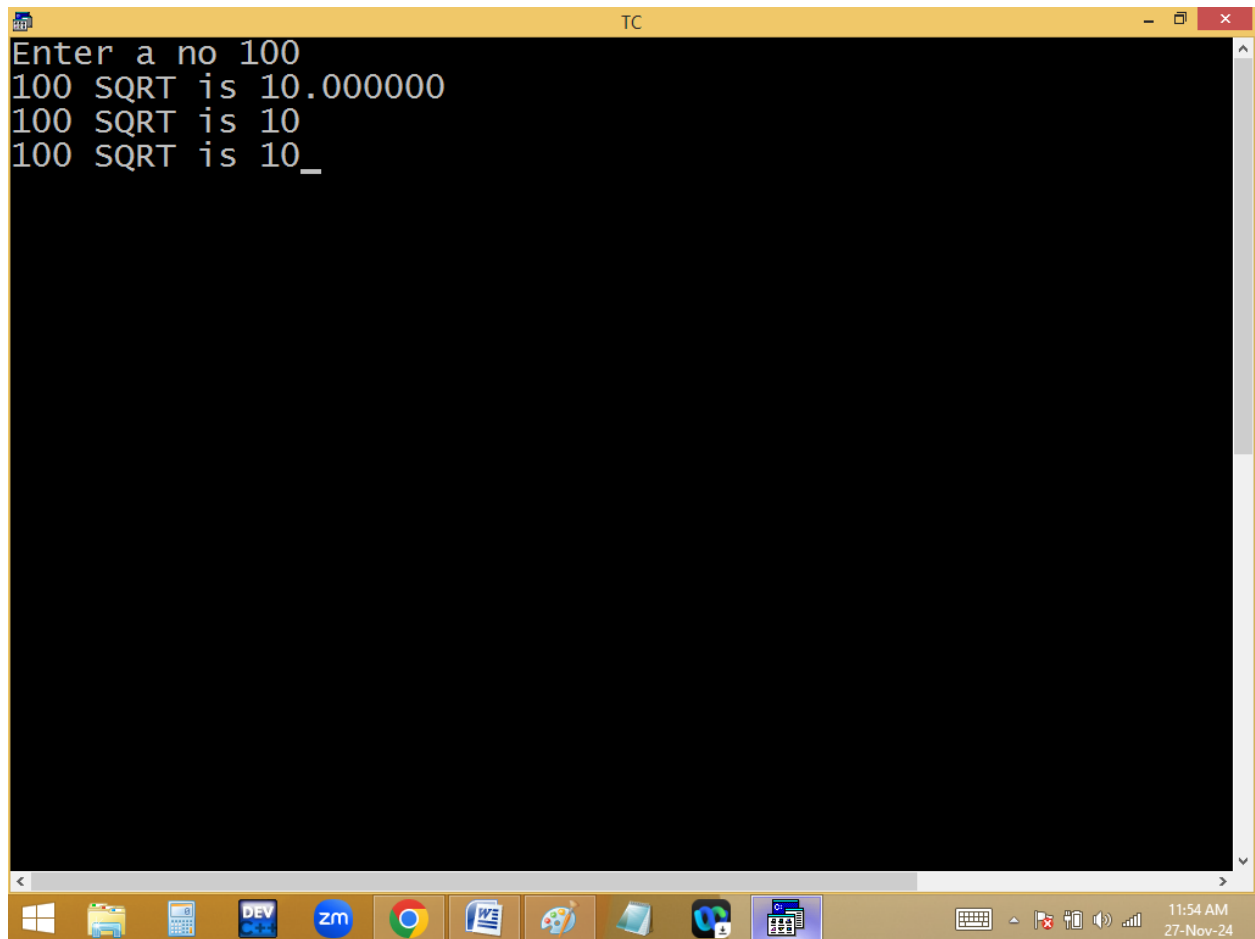
$$\text{SQRT}(100) \rightarrow 10$$



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 10", "Col 26", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The code editor has a blue background and contains the following C program:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
printf("%d SQRT is %f\n",n, sqrt(n));
printf("%d SQRT is %.0f\n",n, sqrt(n));
printf("%d SQRT is %d",n, (int)sqrt(n));
getch();
}
```

The fourth line of the code, `printf("%d SQRT is %d",n, (int)sqrt(n));`, is highlighted in light blue. The bottom status bar shows function key shortcuts: "F1 Help", "F5 Zoom", "F6 Switch", "F7 Trace", "F8 Stop", "F9 Make", and "F10". The Windows taskbar at the bottom includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, a folder, VS Code, and a task view icon. The system clock on the right shows "11:54 AM" and "27-Nov-24".



```
Enter a no 100
100 Sqrt is 10.000000
100 Sqrt is 10
100 Sqrt is 10_
```

Read a baby age in no of days and find the baby age in years, months, weeks and days.

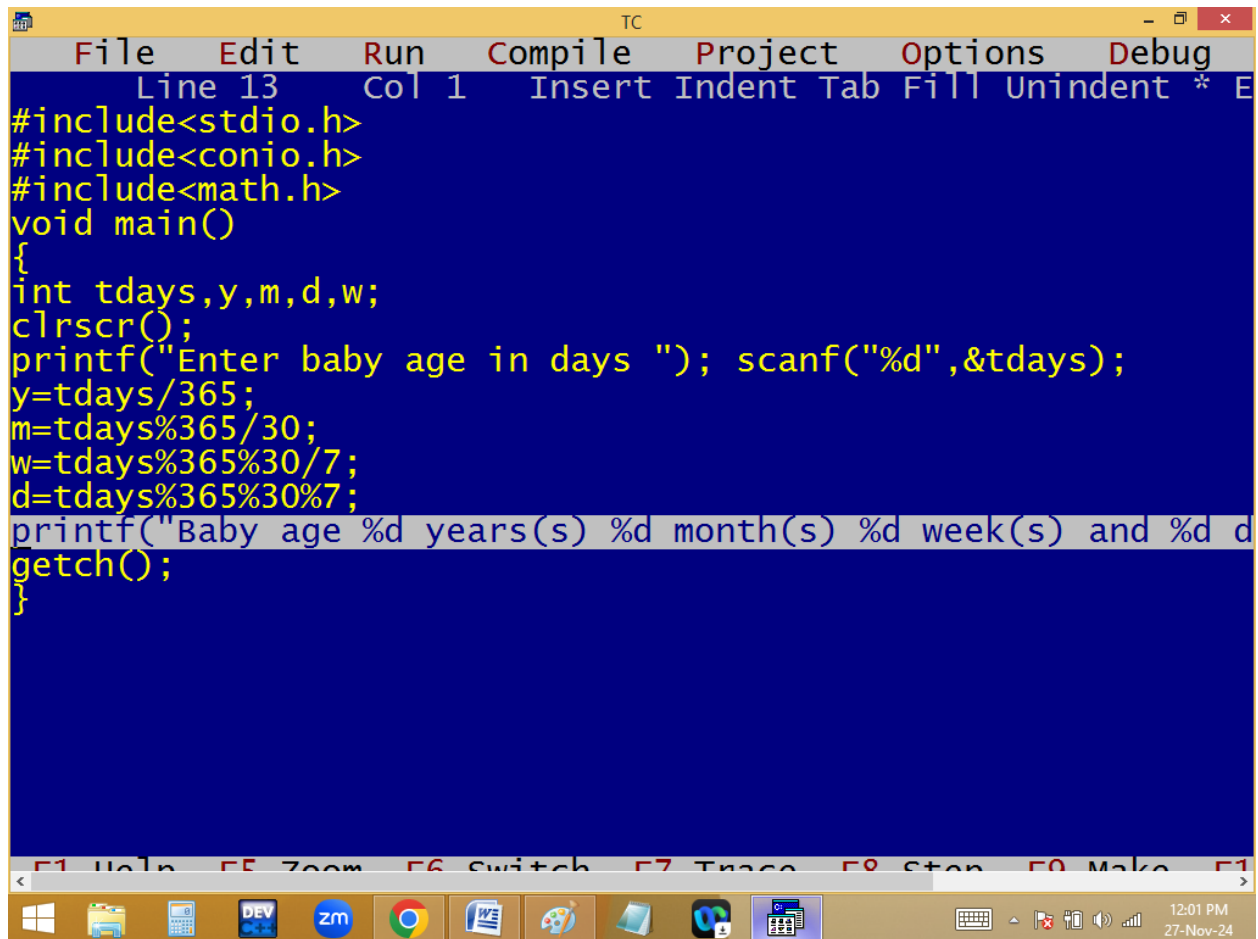
$$y = 500/365=1$$

$$m = 500\%365=135/30=4$$

$$w = 500\%365=135\%30=15/7=2$$

$$d = 500\%365=135\%30=15\%7=1$$

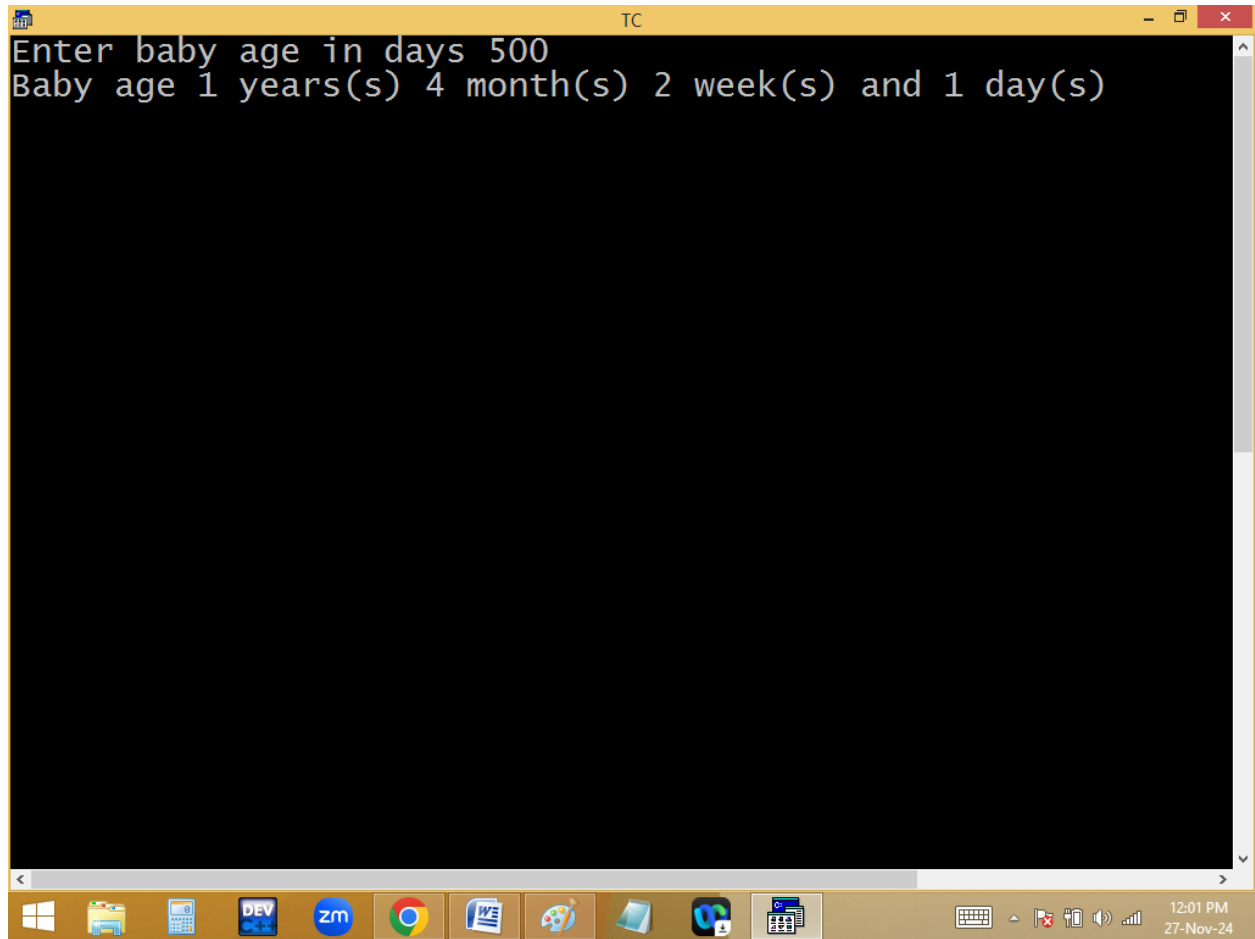
$$\begin{array}{r}
 \text{tdays} \\
 \hline
 365 \overline{) 500} \quad (1 - Y \\
 \underline{365} \\
 30 \overline{) 135} \quad (4 - M \\
 \underline{120} \\
 7 \overline{) 15} \quad (2 - W \\
 \underline{14} \\
 1 - D
 \end{array}$$

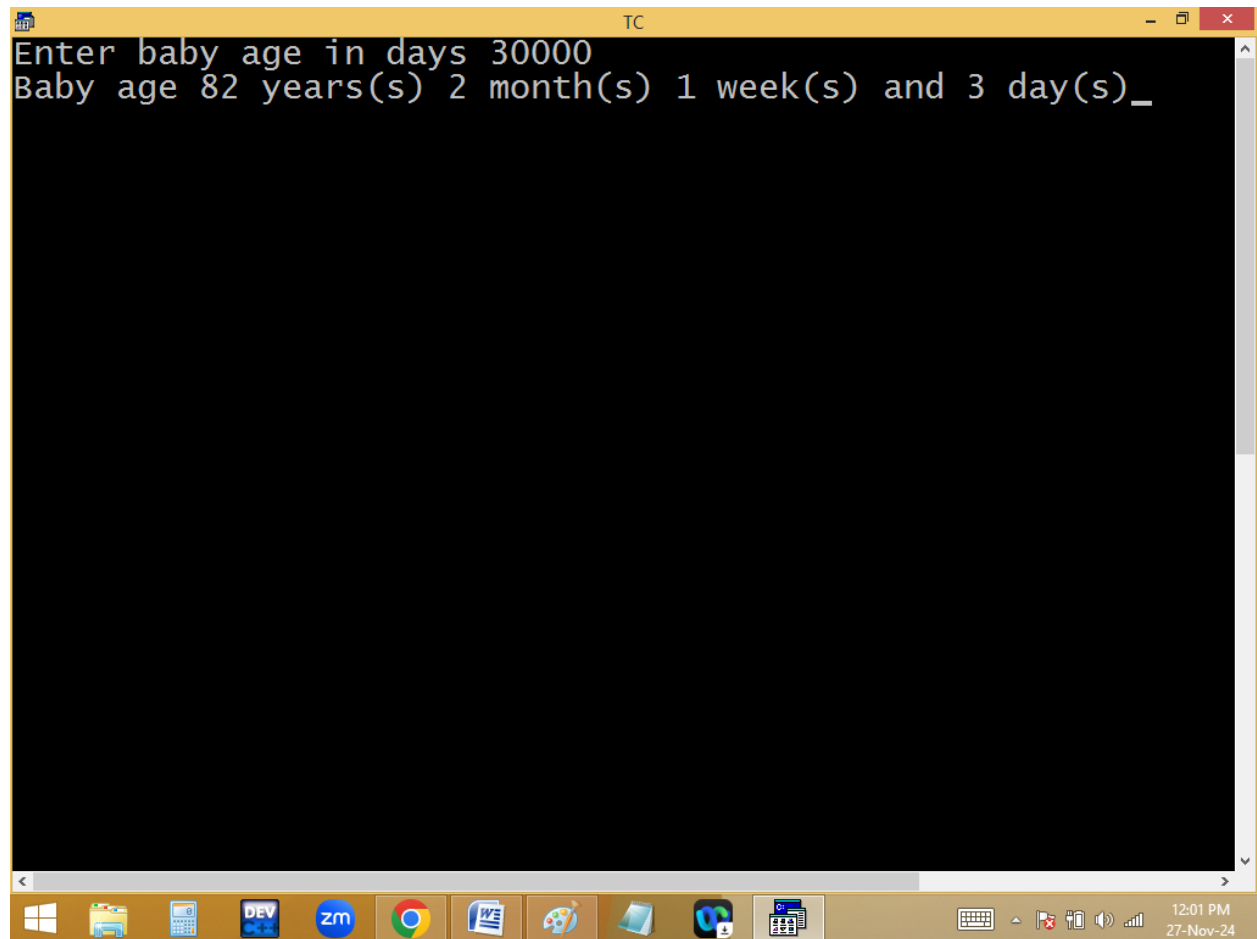


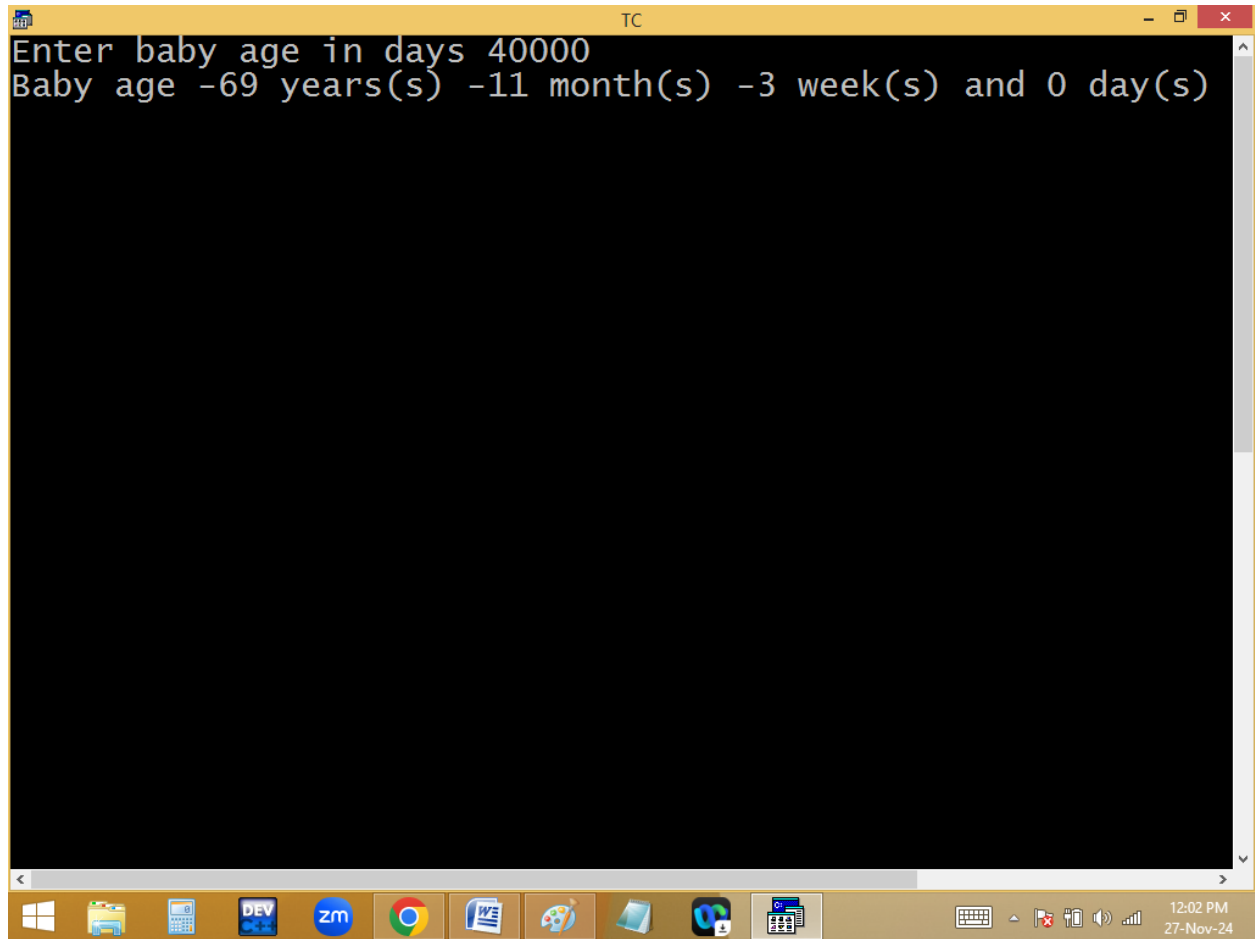
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 13 Col 1" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a dark blue background with yellow text. The code is as follows:

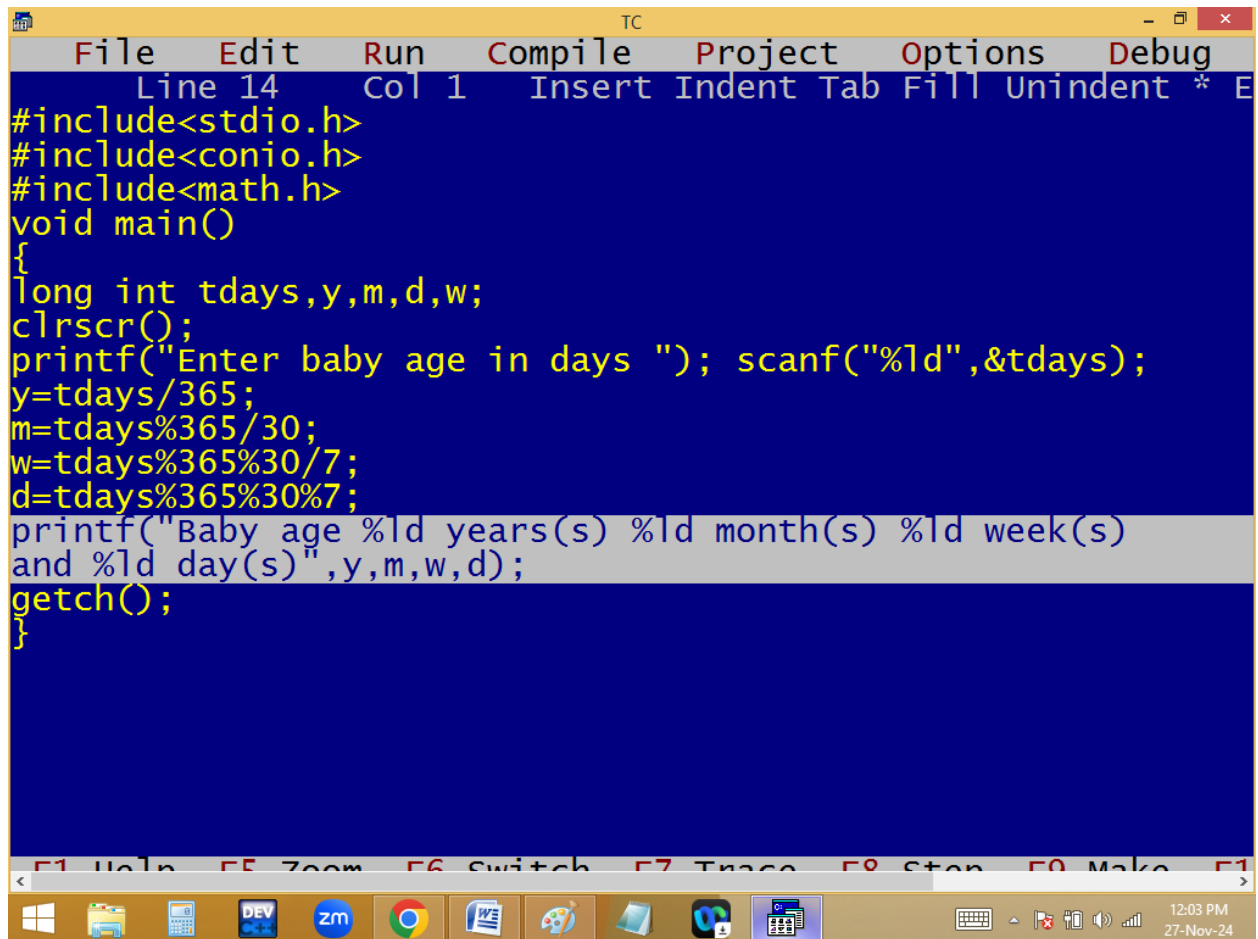
```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int tdays,y,m,d,w;
clrscr();
printf("Enter baby age in days "); scanf("%d",&tdays);
y=tdays/365;
m=tdays%365/30;
w=tdays%365%30/7;
d=tdays%365%30%7;
printf("Baby age %d years(s) %d month(s) %d week(s) and %d d
getch();
}
```

At the bottom of the IDE, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. Below the IDE window is the Windows taskbar, which includes icons for Windows, File Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder icon. The system tray on the right shows the time as 12:01 PM and the date as 27-Nov-24.





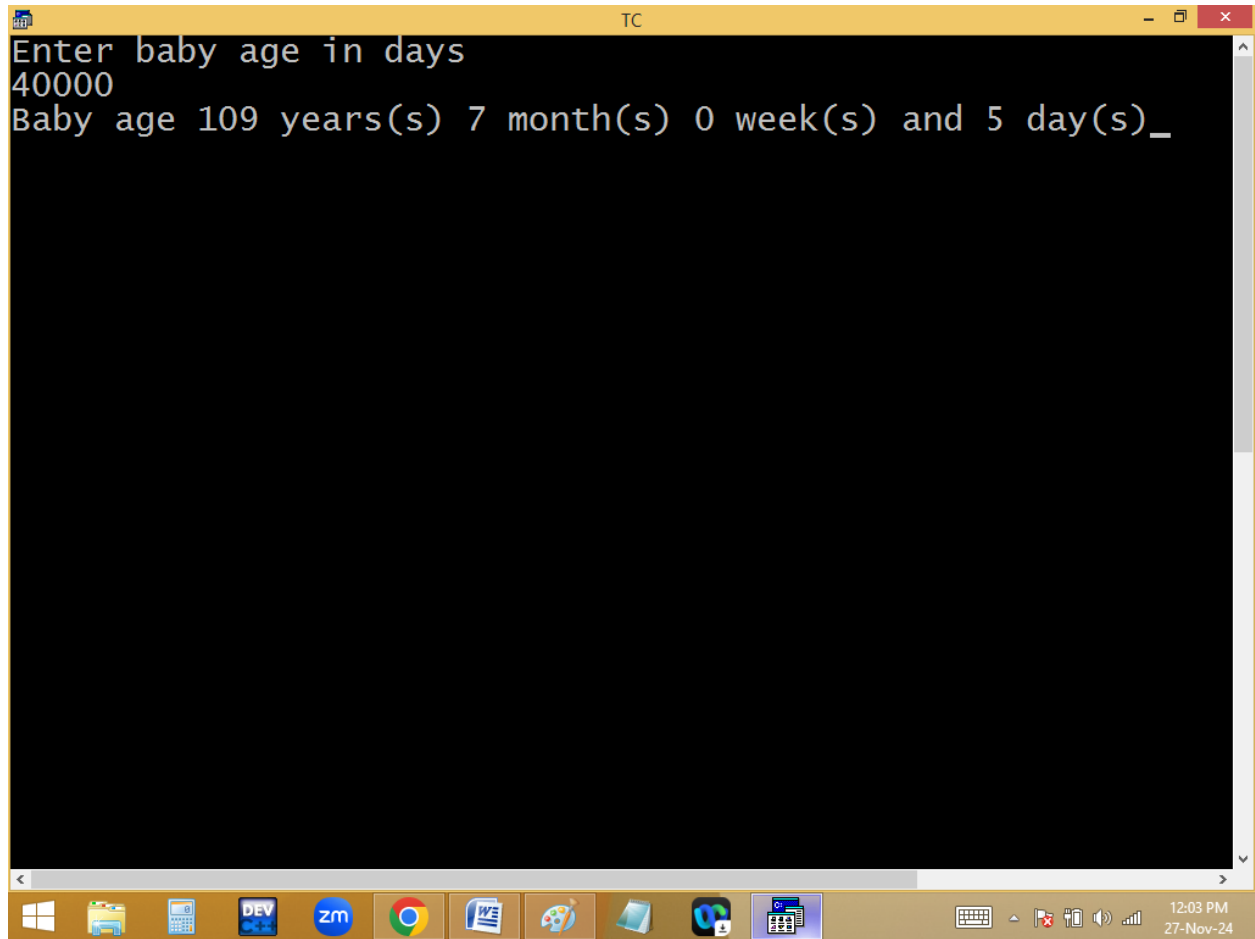


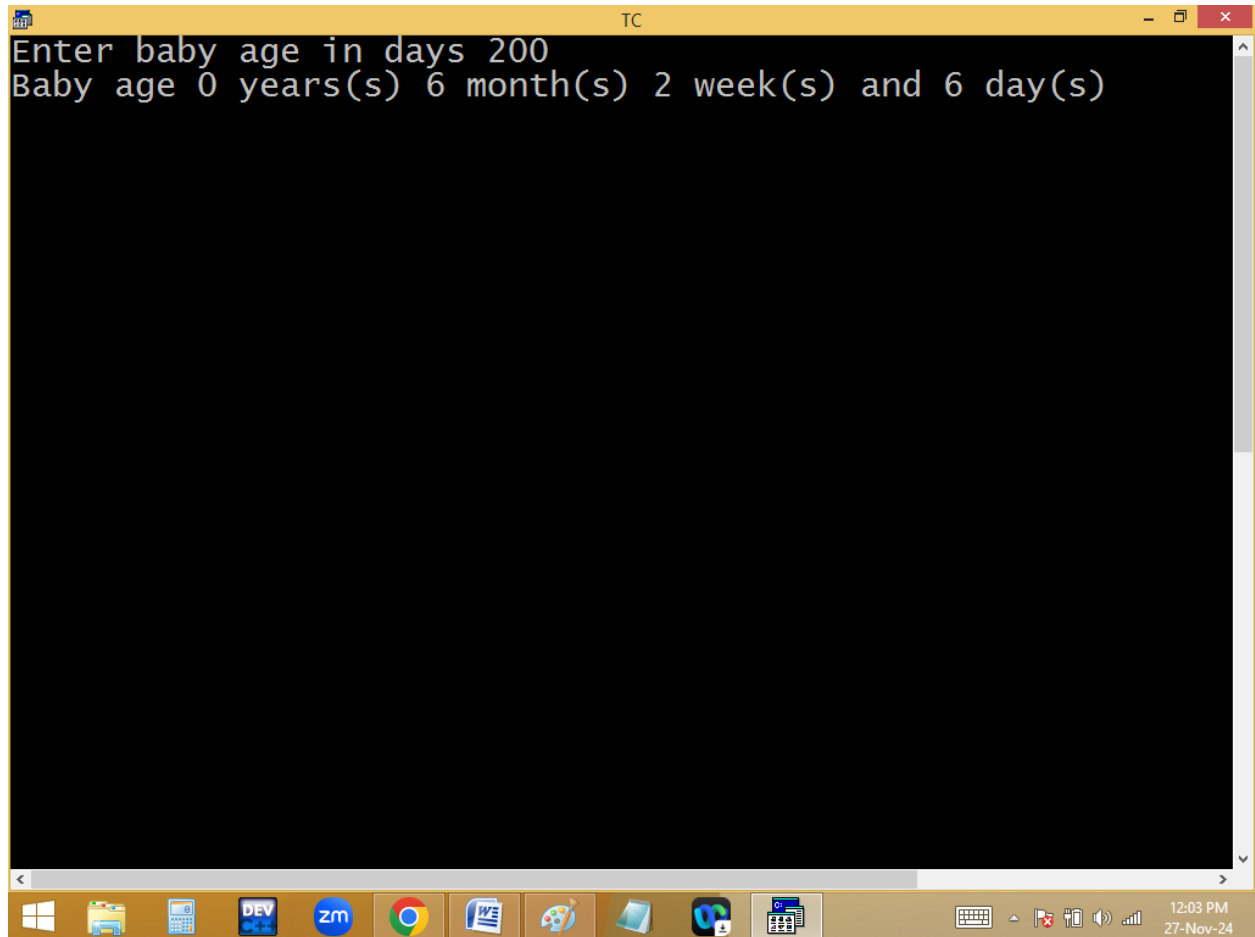


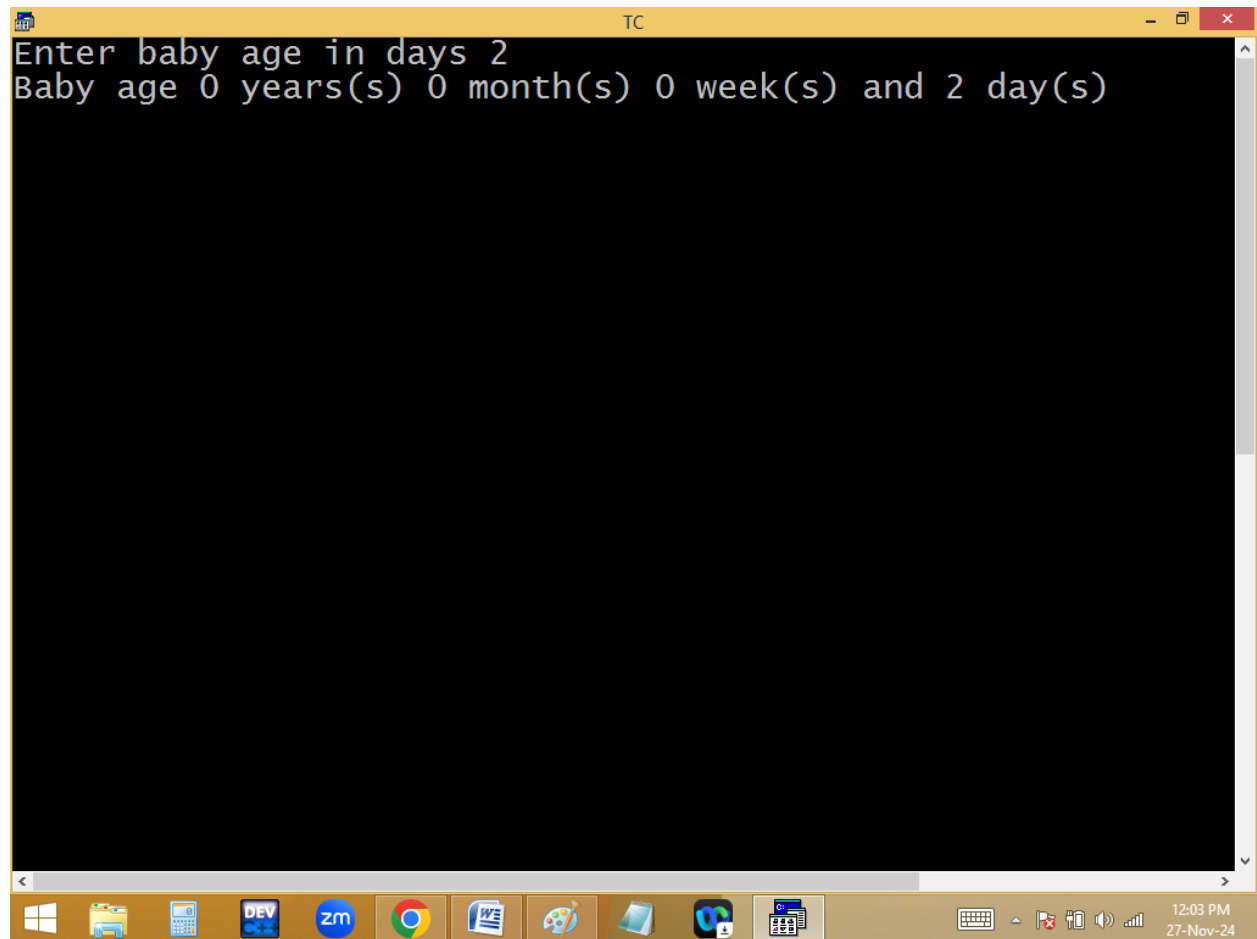
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 14", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
long int tdays,y,m,d,w;
clrscr();
printf("Enter baby age in days "); scanf("%ld",&tdays);
y=tdays/365;
m=tdays%365/30;
w=tdays%365%30/7;
d=tdays%365%30%7;
printf("Baby age %ld years(s) %ld month(s) %ld week(s)
and %ld day(s)",y,m,w,d);
getch();
}
```

At the bottom of the IDE window, there is a toolbar with icons for various functions and a status bar. The status bar shows the time "12:03 PM" and the date "27-Nov-24".





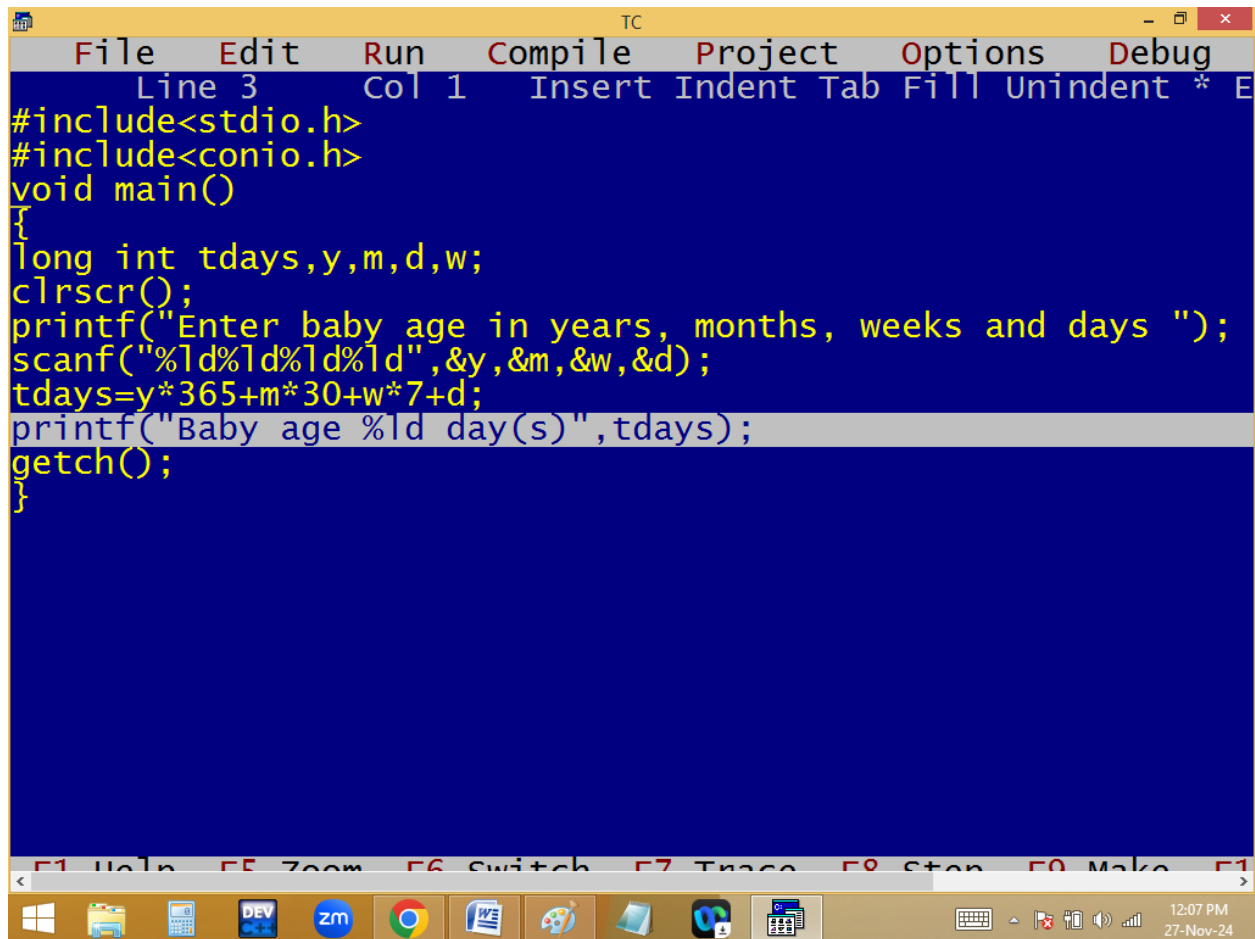


```
TC
Enter baby age in days -3
Baby age 0 year(s) 0 month(s) 0 week(s) and -3 day(s)
```

Read baby age in years, months, weeks and days. Find baby age in days.

$$1y + 4m + 2w + 1d = 500 \text{ days}$$

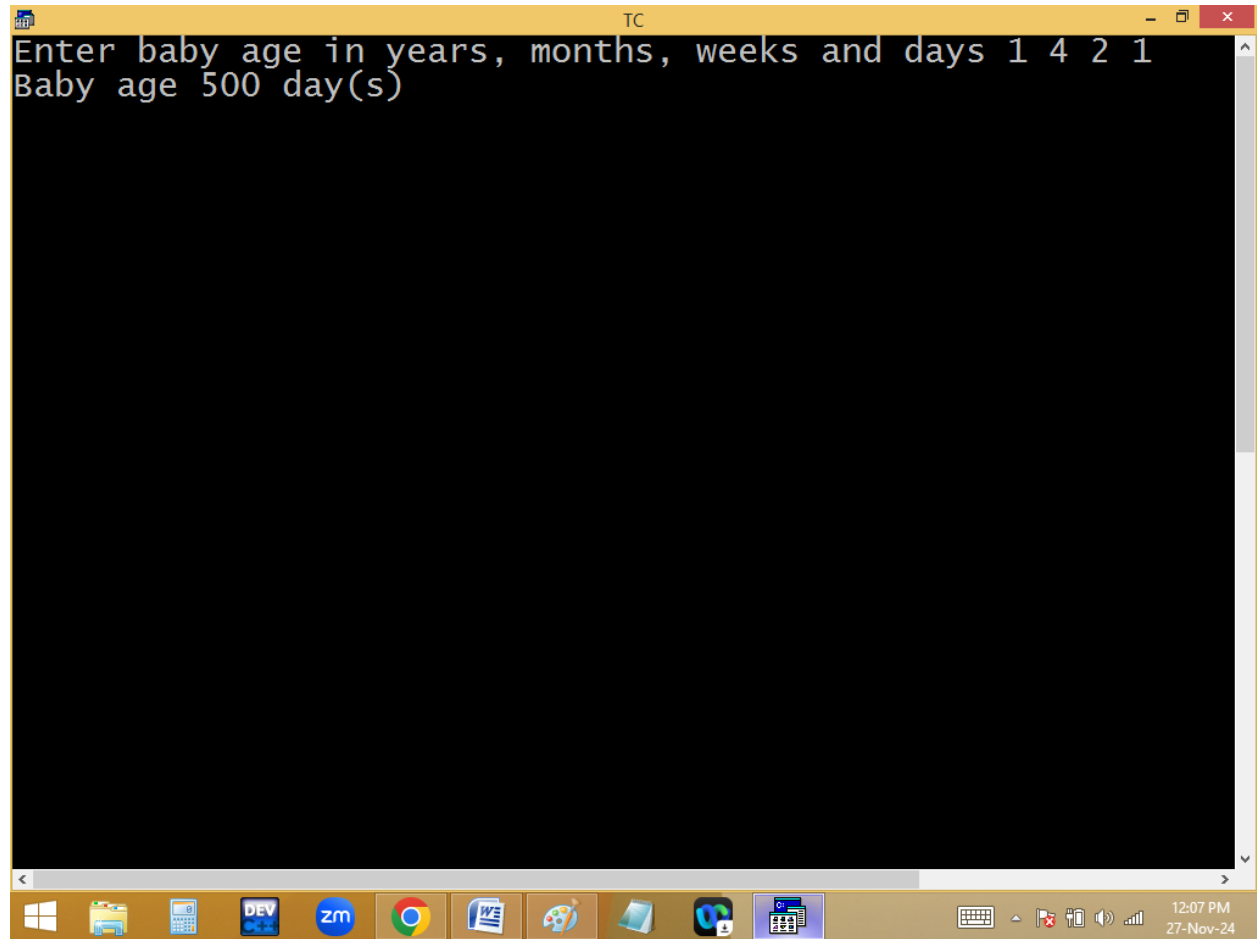
$$1*365 + 4*30 + 2*7 + 1 = 500$$



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 3 Col 1" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long int tdays,y,m,d,w;
    clrscr();
    printf("Enter baby age in years, months, weeks and days ");
    scanf("%ld%ld%ld%ld",&y,&m,&w,&d);
    tdays=y*365+m*30+w*7+d;
    printf("Baby age %ld day(s)",tdays);
    getch();
}
```

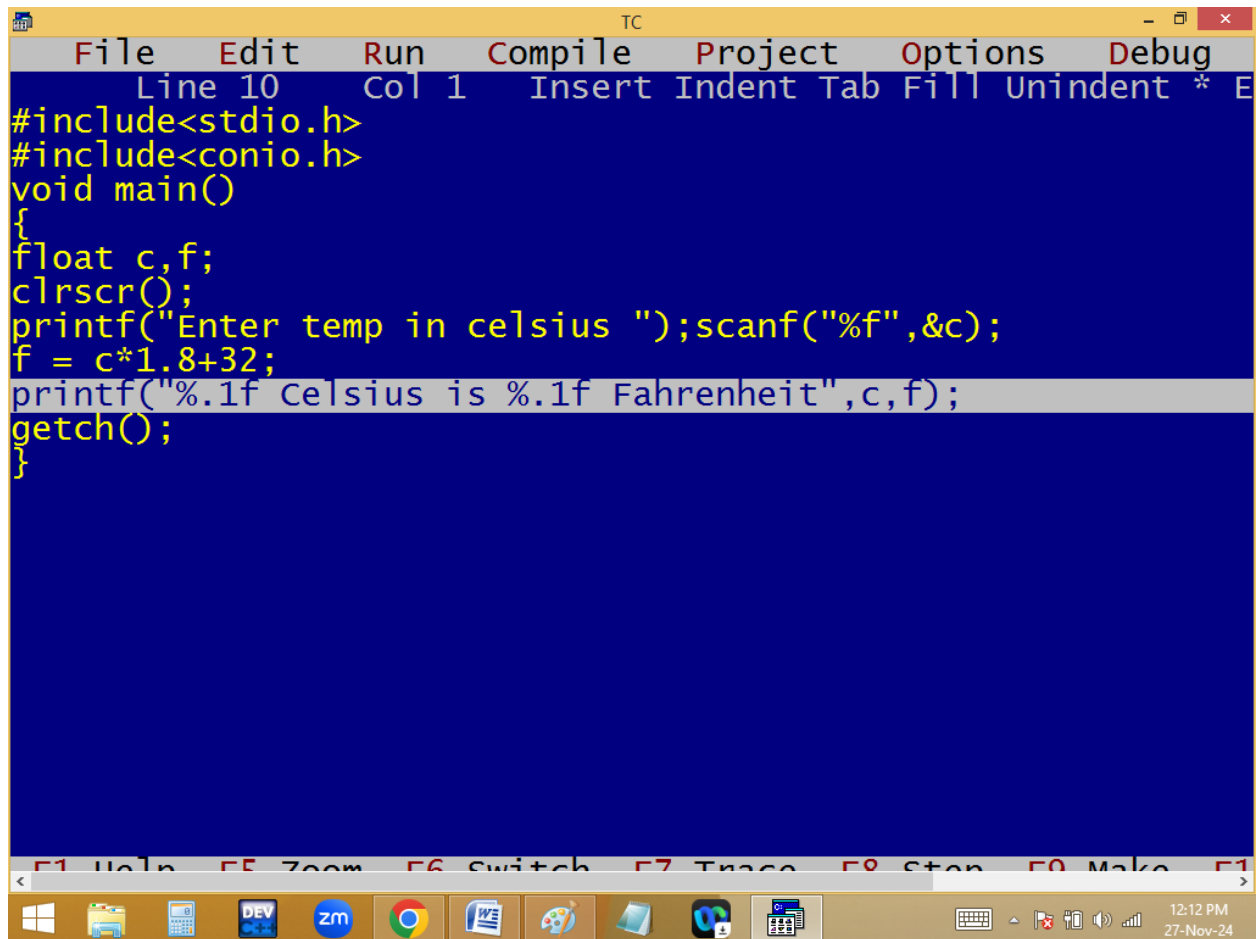
Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner shows "12:07 PM 27-Nov-24".



```
TC
Enter baby age in years, months, weeks and days 23 7 3 2
Baby age 8628 day(s)
```

Celsius to Fahrenheit conversion:

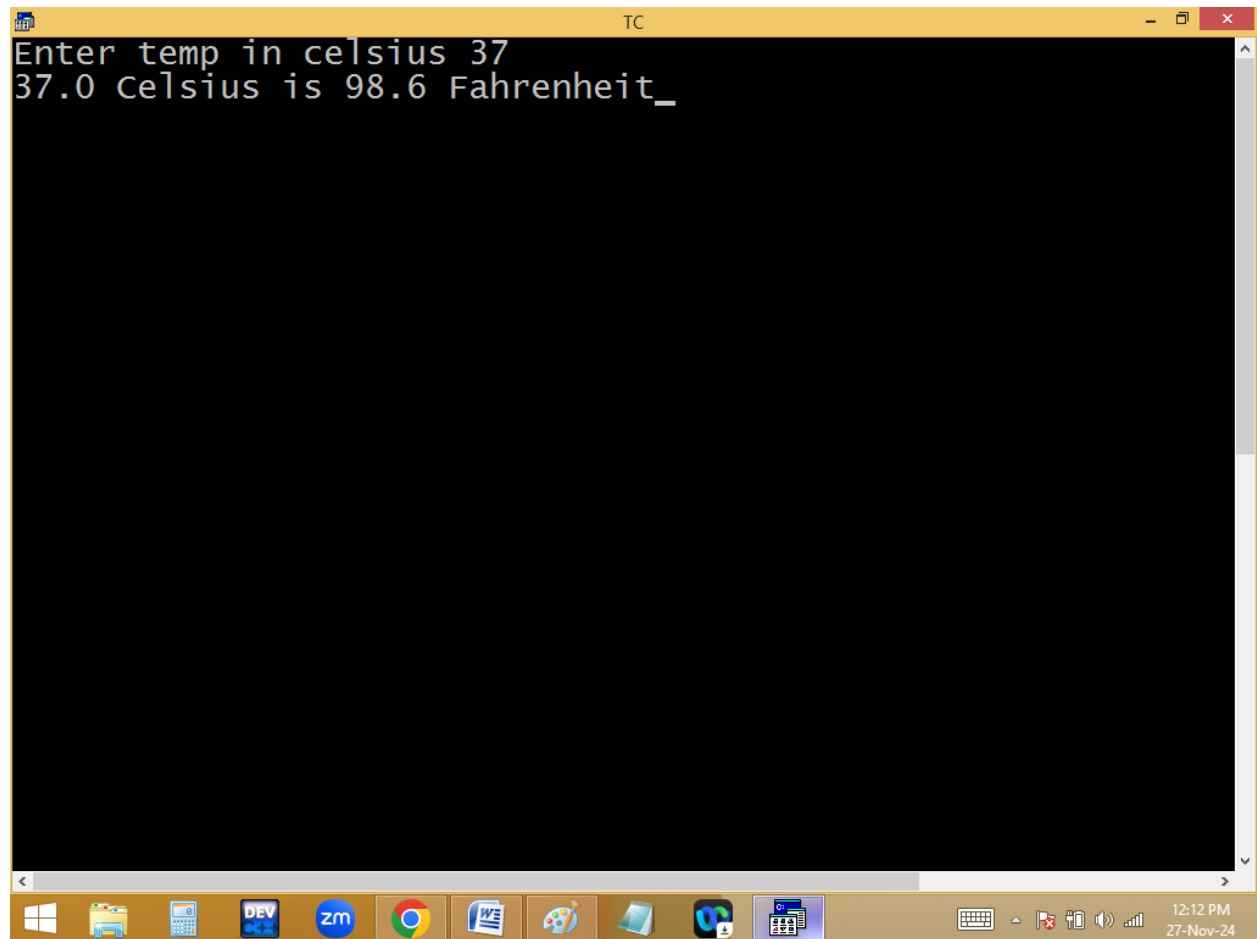
$$f = c * 1.8 + 32$$

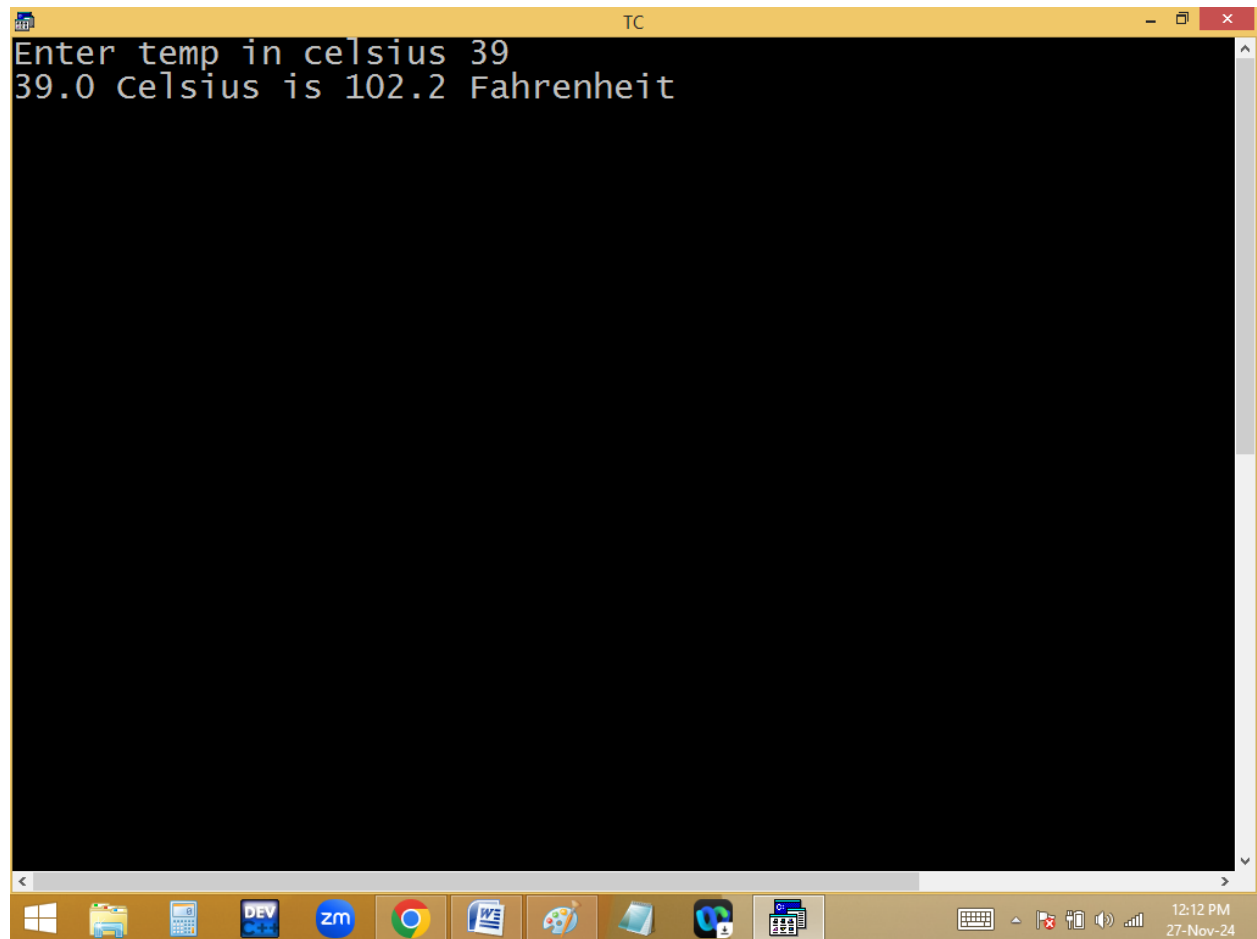


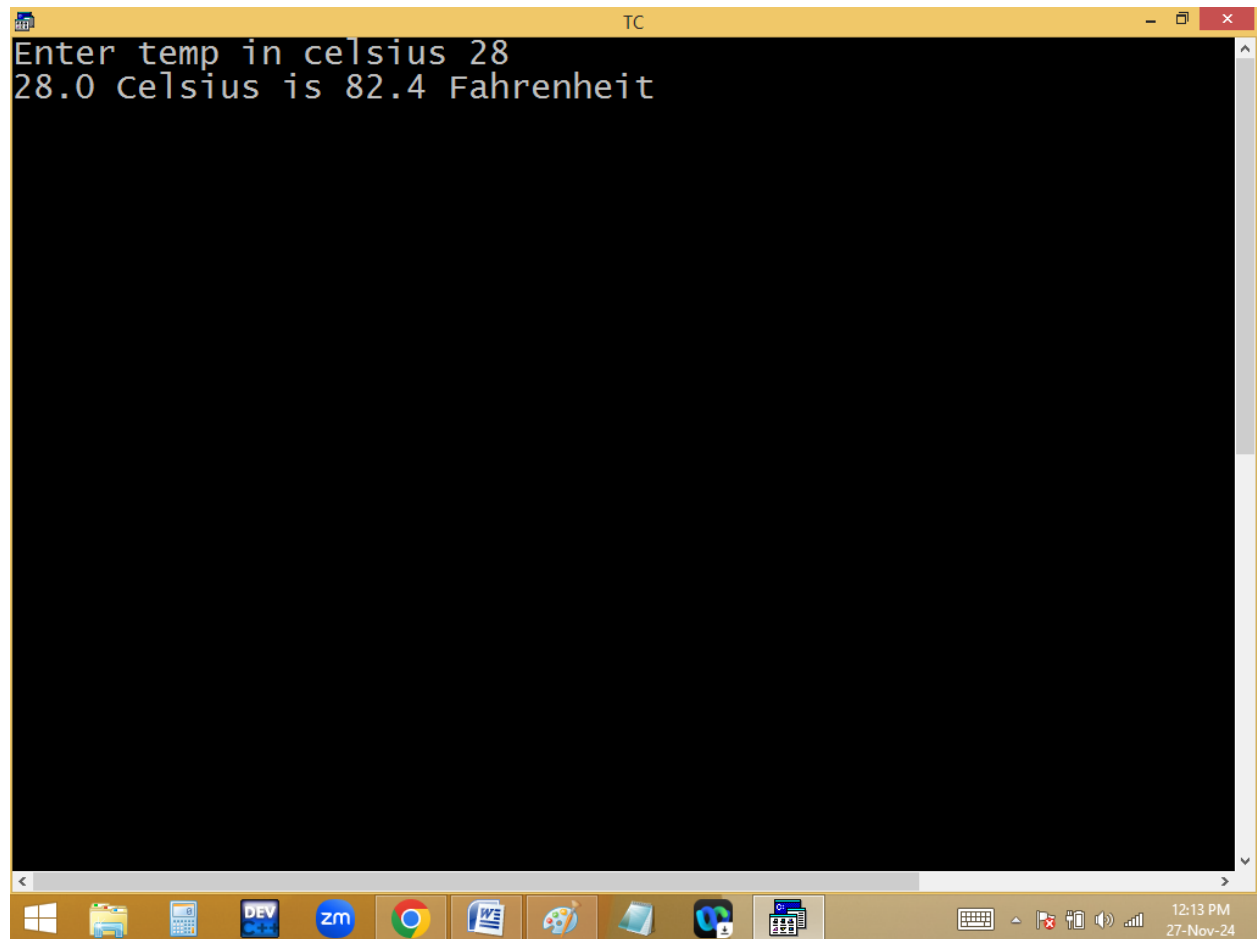
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Edit" menu, there is a submenu with the following options: Line 10, Col 1, Insert, Indent, Tab, Fill, Unindent, and * E. The main editing area has a dark blue background with yellow text. The code is as follows:

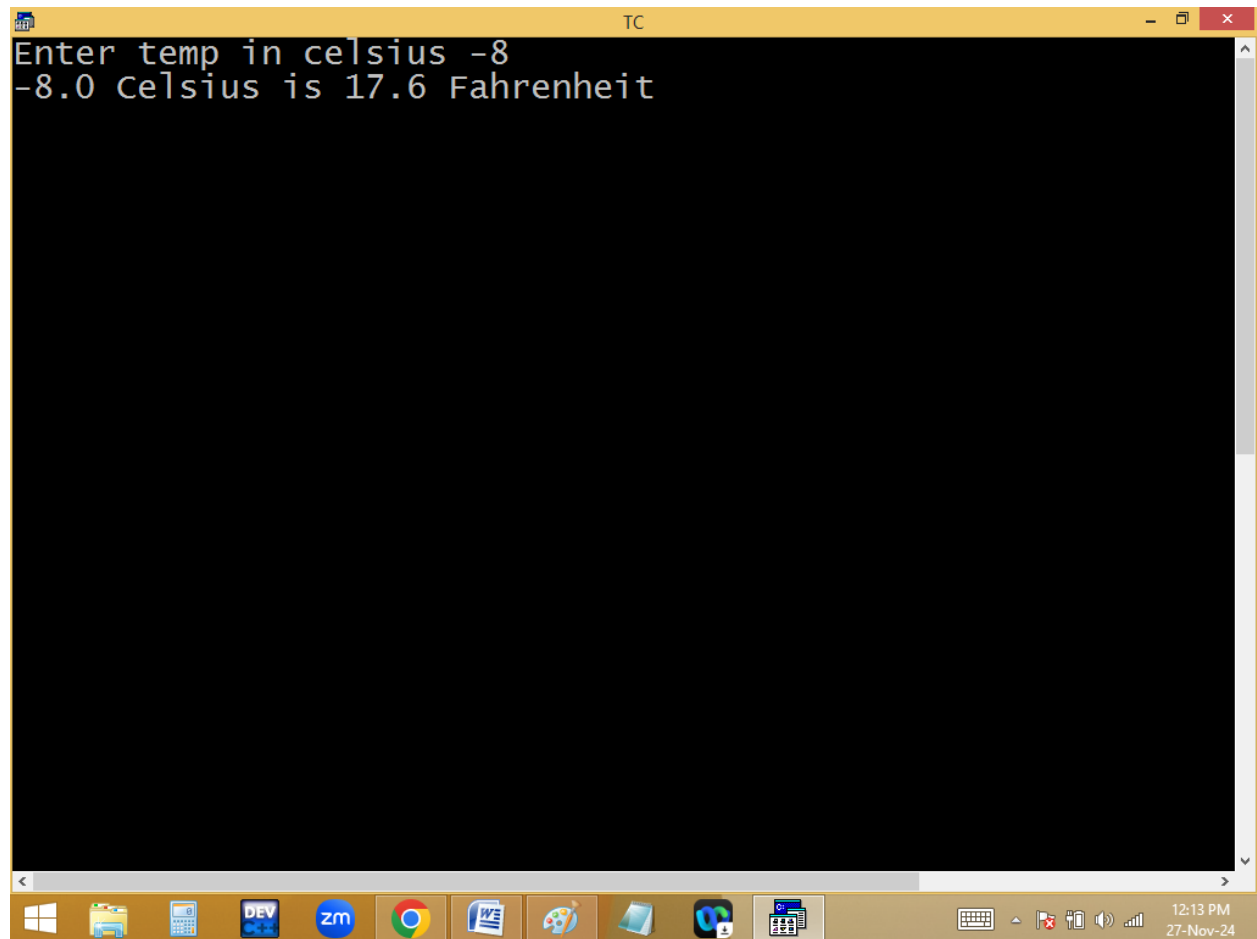
```
#include<stdio.h>
#include<conio.h>
void main()
{
float c,f;
clrscr();
printf("Enter temp in celsius ");scanf("%f",&c);
f = c*1.8+32;
printf("%.1f Celsius is %.1f Fahrenheit",c,f);
getch();
}
```

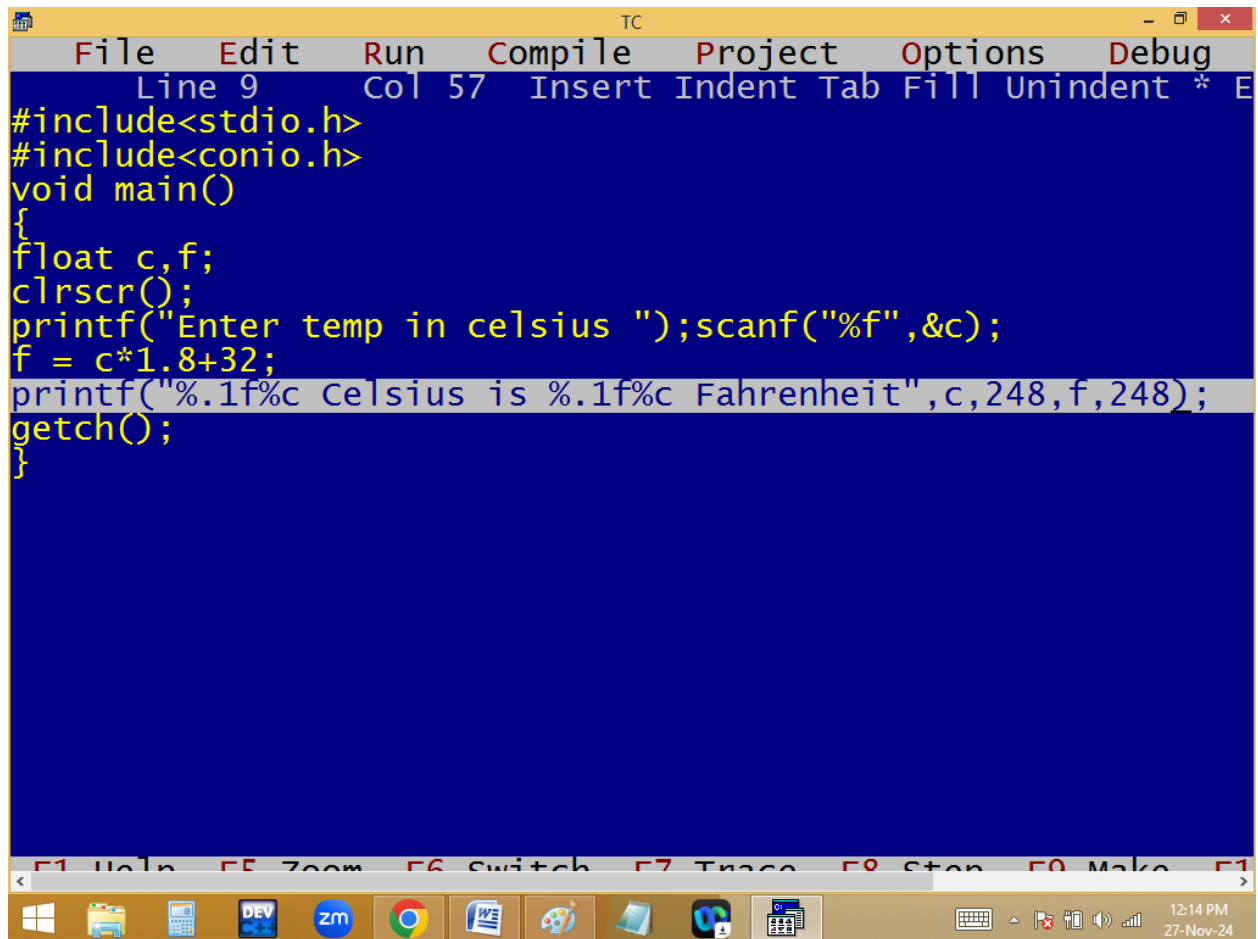
At the bottom of the window, there is a status bar with the following text: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. Below the status bar is a Windows taskbar with various icons, including the Start button, File Explorer, Calculator, DEV, zm, Google Chrome, Word, Paint, and a folder icon. The system clock in the bottom right corner shows the time as 12:12 PM and the date as 27-Nov-24.











The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 9", "Col 57", and "Insert Indent Tab Fill Unindent * E". The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float c,f;
clrscr();
printf("Enter temp in celsius ");scanf("%f",&c);
f = c*1.8+32;
printf("%.1f%c Celsius is %.1f%c Fahrenheit",c,248,f,248);
getch();
}
```

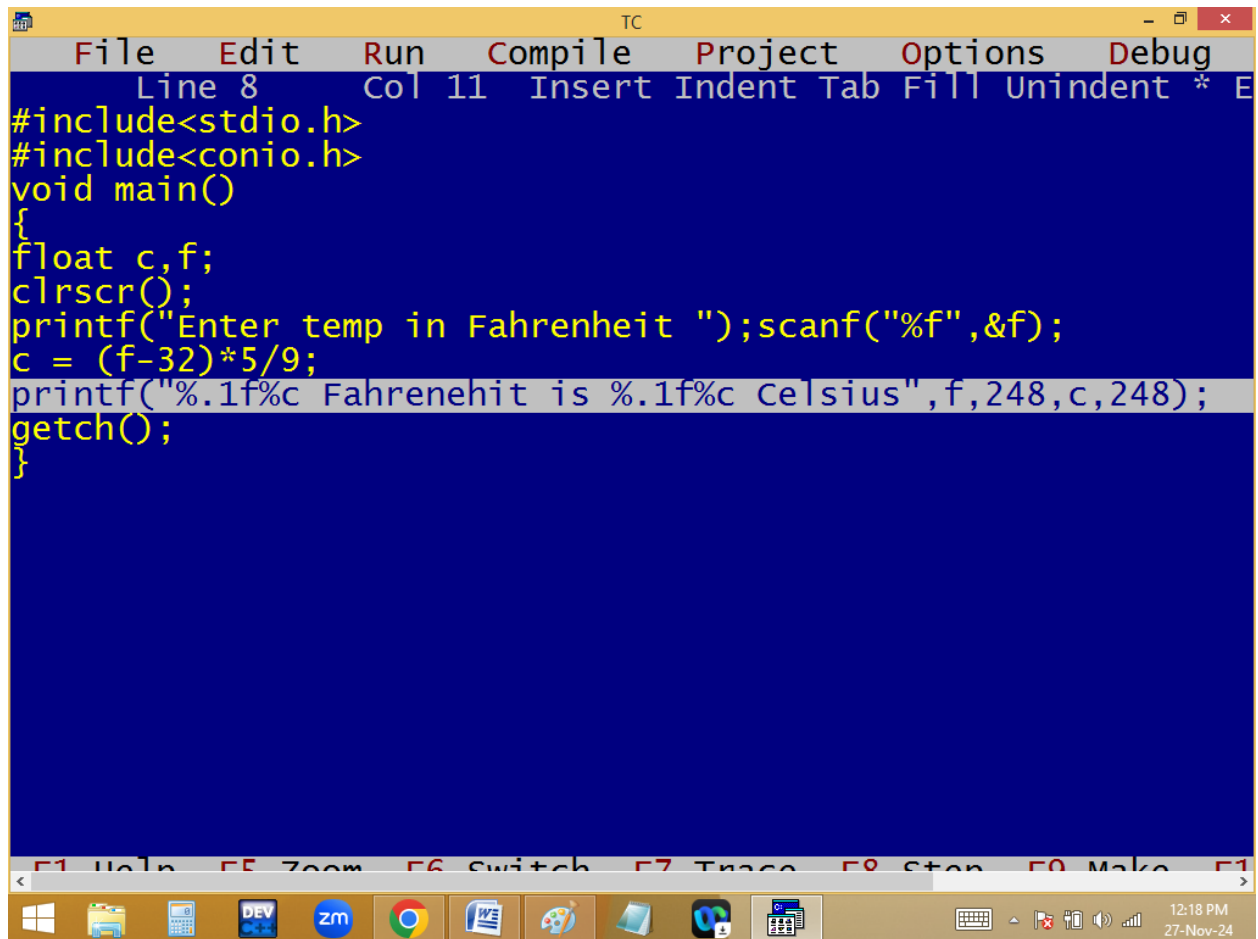
The bottom status bar shows function key shortcuts: "F1 Help", "F5 Zoom", "F6 Switch", "F7 Trace", "F8 Stop", "F9 Make", and "F10". The Windows taskbar at the bottom includes icons for Windows, File Explorer, Calculator, DEV C++, ZOOM, Google Chrome, Word, Paint, and a folder. The system clock shows "12:14 PM" and "27-Nov-24".

```
TC
Enter temp in celsius 37
37.0° Celsius is 98.6° Fahrenheit
```

^{37.0 °} ^{98.6 °}
`p("%.1f%c cel is %.1f%c Fahrenheit",c,248,f,248);`

Fahrenheit to Celsius:

$$C = f - 32 * 5/9$$



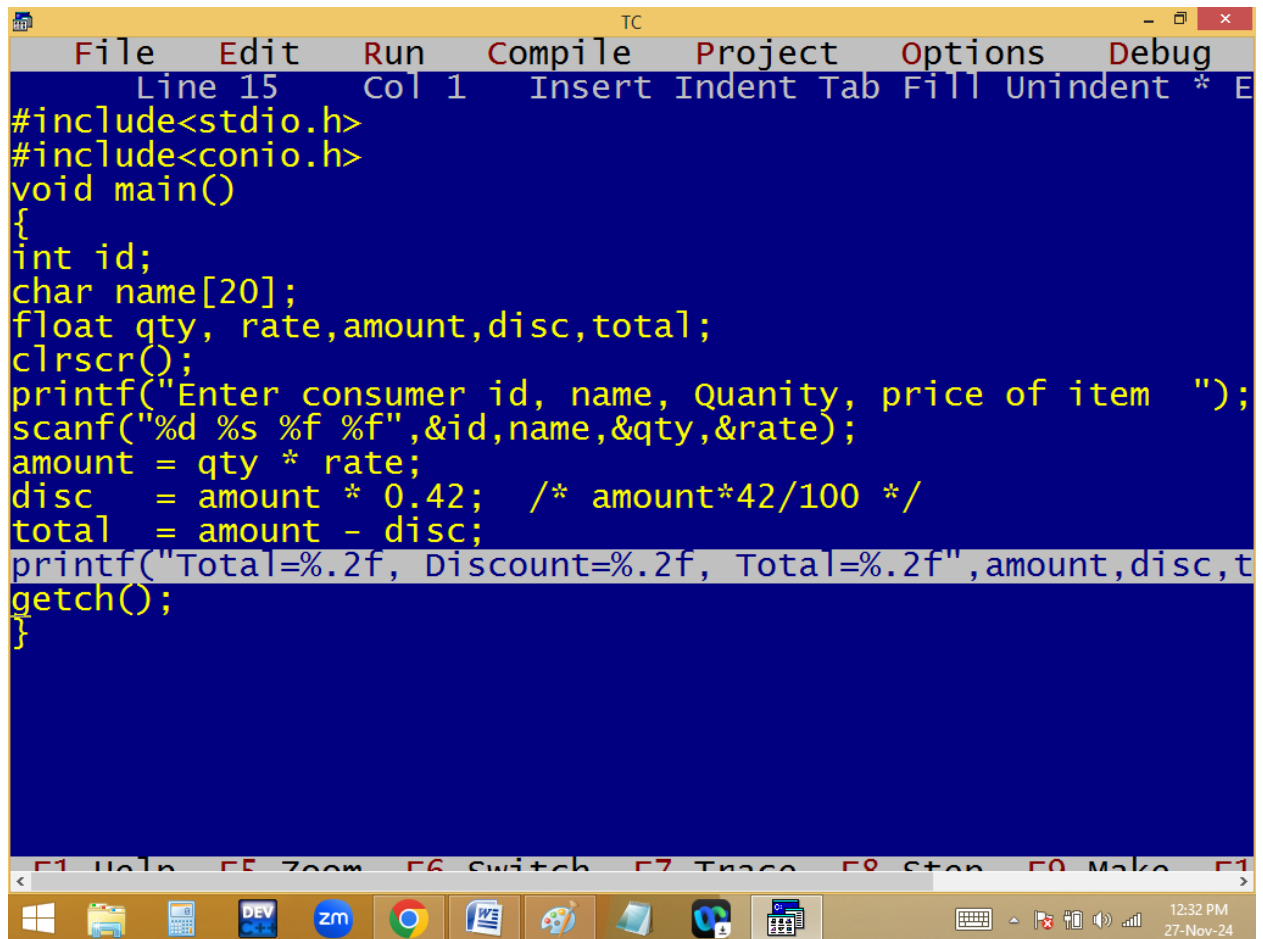
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar indicates "Line 8" and "Col 11". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
float c,f;
clrscr();
printf("Enter temp in Fahrenheit ");scanf("%f",&f);
c = (f-32)*5/9;
printf("%.1f%c Fahrenehit is %.1f%c Celsius",f,248,c,248);
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "12:18 PM" and the date "27-Nov-24".

```
TC
Enter temp in Fahrenheit 98.6
98.6° Fahrenehit is 37.0° Celsius_
```

Read a consumer id, name, Quantity purchased and rate of item. Find the amount, 42% discount and total.



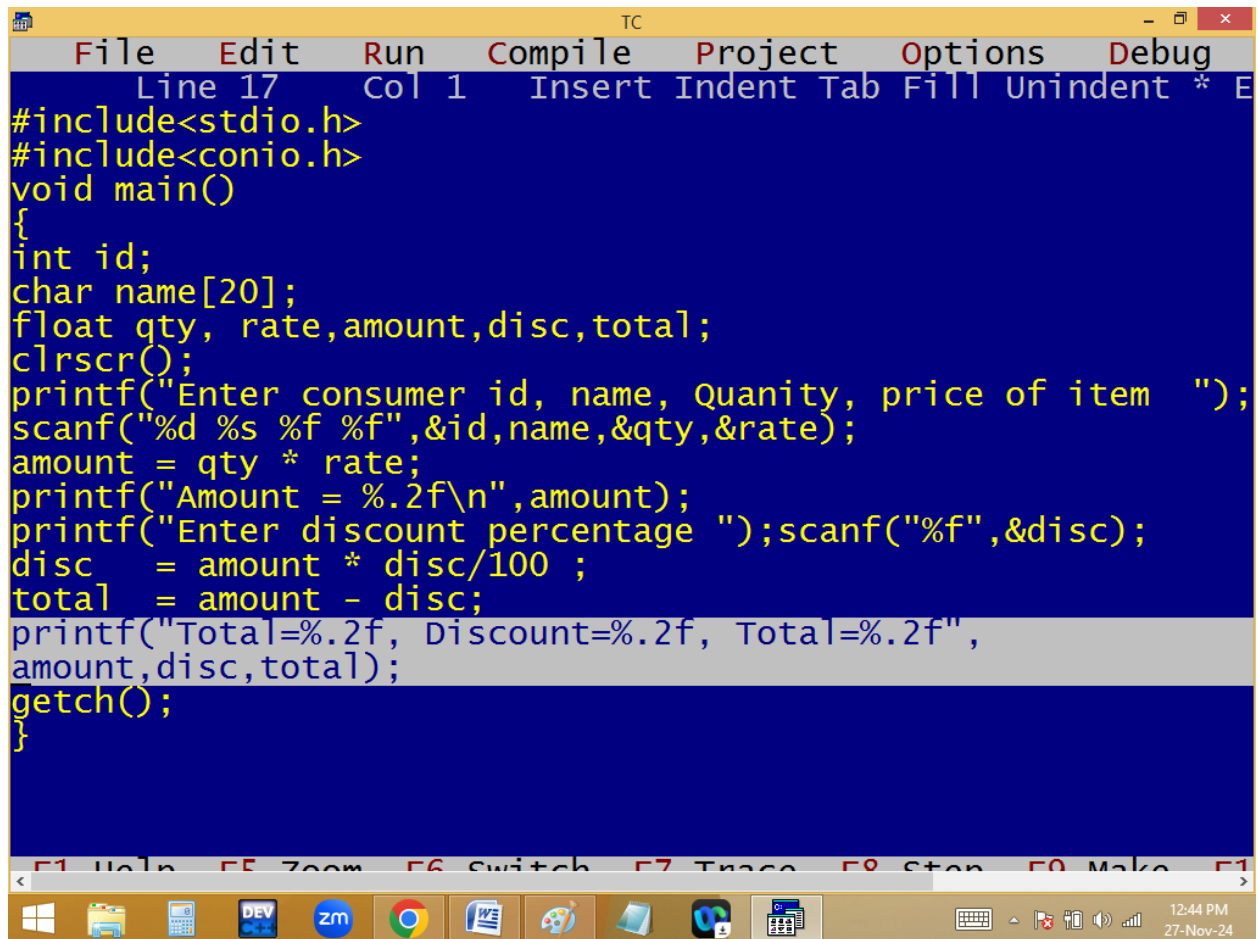
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 15 Col 1" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The code is written in yellow text on a dark blue background. It includes headers for `stdio.h` and `conio.h`, and defines a `main` function. The program prompts the user to enter consumer details, calculates the amount, applies a 42% discount, and prints the results. The last line of code is highlighted in light blue. The bottom status bar shows function key shortcuts (F1-F11) and the system clock (12:32 PM, 27-Nov-24).

```
#include<stdio.h>
#include<conio.h>
void main()
{
int id;
char name[20];
float qty, rate,amount,disc,total;
clrscr();
printf("Enter consumer id, name, Quantity, price of item ");
scanf("%d %s %f %f",&id,name,&qty,&rate);
amount = qty * rate;
disc  = amount * 0.42; /* amount*42/100 */
total  = amount - disc;
printf("Total=%.2f, Discount=%.2f, Total=%.2f",amount,disc,t
getch();
}
```

```
TC
Enter consumer id, name, Quantity, price of item
1 bablu 1 10000
Total=10000.00, Discount=4200.00, Total=5800.00_
```

```
TC
Enter consumer id, name, Quantity, price of item
2 kumari 1 1
Total=1.00, Discount=0.42, Total=0.58
```

Dynamic [instant]discount:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 17", "Col 1", and "Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes headers for `<stdio.h>` and `<conio.h>`, and defines a `main()` function. The program prompts the user to enter consumer details, calculates the amount, applies a discount, and prints the final total. The bottom status bar shows function key shortcuts (F1-F11) and the system clock (12:44 PM, 27-Nov-24).

```
#include<stdio.h>
#include<conio.h>
void main()
{
int id;
char name[20];
float qty, rate,amount,disc,total;
clrscr();
printf("Enter consumer id, name, Quantity, price of item ");
scanf("%d %s %f %f",&id,name,&qty,&rate);
amount = qty * rate;
printf("Amount = %.2f\n",amount);
printf("Enter discount percentage ");scanf("%f",&disc);
disc  = amount * disc/100 ;
total  = amount - disc;
printf("Total=%.2f, Discount=%.2f, Total=%.2f",
amount,disc,total);
getch();
}
```

```
TC
Enter consumer id, name, Quantity, price of item
1 bablu 5 20
Amount = 100.00
Enter discount percentage 0
Total=100.00, Discount=0.00, Total=100.00
```

```
TC
Enter consumer id, name, Quantity, price of item
2 priya 1 75000
Amount = 75000.00
Enter discount percentage 60
Total=75000.00, Discount=45000.00, Total=30000.00_
```

```
TC
Enter consumer id, name, Quantity, price of item 3
jack 10 200
Amount = 2000.00
Enter discount percentage 5
Total=2000.00, Discount=100.00, Total=1900.00_
```

